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Original Article

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Primitive Arc Magmatism and Delamination: Petrology and Geochemistry of Pyroxenites from the Cabo Ortegal Complex, Spain

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ABSTRACT

The genesis of primitive arc magmas has had a major impact on continent formation through time, but the rarity of exposures of deep arc sections limits our understanding of the details of melt migration and differentiation. Abundant pyroxenites are exposed within a 600 m thick section of arc-related mantle harzburgites and dunites in the Herbeira massif of the Cabo Ortegal Complex, Spain. We report a combination of field and petrographic observations with *in situ* and whole-rock geochemical studies of these pyroxenites. After constraining the effects of secondary processes (serpentinization, fluid or melt percolation and amphibolitization), we determine that the low Al content of pyroxenes, high abundance of compatible elements and the absence of plagioclase reflect melt–peridotite interaction and crystal segregation from primitive hydrous melts at relatively low pressure ($<1{\cdot}2$ GPa). Olivine clinopyroxenites and olivine websterites preserving dunite lenses (type 1 and 3 pyroxenites) represent the products of partial replacement of peridotites at decreasing melt/rock ratio following the intrusion of picritic melts. Massive websterites (type 2) may represent the final products of this reaction at higher melt/rock ratios. They crystallized from more Si-rich (boninitic) melts, potentially generated through differentiation of the initially picritic melts or intruded as dykes and veins. Rare opx-rich websterites (type 4) were produced by interaction of these melts with dunites. Chromatographic re-equilibration accompanied late-magmatic crystallization of amphibole from migrating or trapped residual melts. This percolative fractional crystallization produced a range of rare earth element (REE) patterns from spoon-shaped in type 1 pyroxenites to strongly light REE (LREE)-enriched in type 2 and 3 pyroxenites. Particularly high CaO/Al_2O_3 ratios (2·2–11·3) and the selective enrichment of large ion lithophile elements (LILE) over high field strength elements (HFSE) in Cabo Ortegal pyroxenites suggest the generation of Ca-rich picritic–boninitic parental melts via low-degree, second-stage melting of a refractory lherzolite at <2 GPa, following percolation of slab-derived fluids and/or carbonatite melts. Pyroxenites and their host peridotites record high-temperature deformation followed by the development of sheath folds and mylonites. Peak metamorphism was then reached under eclogite-facies conditions (1·6–1·8 GPa and 780–800°C) as recorded by undeformed garnet coronas around spinel. We suggest that this episode corresponds to the delamination of an arc root owing to gravitational instabilities arising from the presence of abundant pyroxenites within mantle harzburgites. Retrograde metamorphism and hydration under amphibolite-facies conditions were recorded by abundant post-kinematic amphibole, which corresponds to the exhumation of the arc root after its intrusion into a subduction zone. The Cabo Ortegal Complex thus preserves a unique section of delaminated arc root, providing evidence for the significant role of melt–peridotite interaction during the differentiation of primitive arc magmas at depth.

Key words: northwestern Iberia; Variscan suture; volcanic arc; mantle metasomatism; peritectic reaction; amphibolitization

INTRODUCTION

Mafic lithologies such as pyroxenite and eclogite play a major role in the development of chemical heterogeneity in the upper mantle and in the recycling of elements between the deep Earth and its external envelopes. They contribute to the generation and evolution of mantle-derived melts and it is increasingly apparent that their signature may be recognizable in the source region of mid-ocean ridge and intra-plate basalts (e.g. Hirschmann & Stolper, 1996; Hirschmann *et al*., 2003; Kogiso *et al*., 2003, 2004; Sobolev *et al*., 2005; Lambart *et al*., 2012, 2013). Abundant pyroxenites representing cumulates left after intracrustal fractionation of primitive, mantle-derived arc melts are also expected from experimental studies (Kay & Kay, 1985; DeBari *et al*., 1987; Müntener & Ulmer, 2006). However, in the few exposed sections of arc crust and sub-arc mantle, these are either much less abundant than expected from mass-balance calculations, as in the case of the Talkeetna arc, Alaska (Debari & Sleep, 1991; Miller & Christensen, 1994; Greene *et al*., 2006; Kelemen *et al*., 2014), or their origin as crustal cumulates is debated because of the lack of a crystal fractionation linkage, as in the case of the Kohistan arc, Pakistan (Khan *et al*., 1993; Burg *et al*., 1998; Garrido *et al*., 2007). This contradiction has major implications for our understanding of the nature of primary arc magmas and their differentiation, and thus on the debated role of arc magmatism and density-sorting processes during the generation and refinement of the continental crust (e.g. Kay & Kay, 1988; Rudnick, 1995; Holbrook *et al*., 1999; Kelemen *et al*., 2014; Garrido *et al*., 2006; Hawkesworth & Kemp, 2006; Lee *et al*., 2006; Jagoutz & Kelemen, 2015). In addition, a better understanding of the petrogenesis of arc-related pyroxenites may provide key constraints on the contribution of slab-derived components to mantle-wedge melts in the source of primary arc magmas (Kelemen, 1990; Kostopoulos & Murton, 1992; Hawkesworth *et al*., 1993; Schiano *et al*., 2000; Bénard & Ionov, 2013).

Pyroxenites occur in variable amounts in the crustal section of ophiolites (e.g. Ceuleneer & Le Sueur, 2008; Clénet *et al*., 2010) and frequently in greater proportion in orogenic peridotite massifs (e.g. Kornprobst, 1969; Conquéré, 1977; Fabriès *et al*., 1991; Pearson *et al*., 1993; Hirschmann & Stolper, 1996; Garrido & Bodinier, 1999; Downes, 2007; Gysi *et al*., 2011; Borghini *et al*., 2013, 2016; Bodinier & Godard, 2014). They have been interpreted as crystallized melts produced by partial melting of their host peridotites (Dickey, 1969; Van Calsteren, 1978) or elongated slices of subducted oceanic crust recycled through mantle convection (Polvé & Allègre, 1980; Hamelin & Allègre, 1985; Allègre & Turcotte, 1986; Kornprobst *et al*., 1990) but they are commonly believed to represent minerals segregated from migrating melts in magmatic conduits (O'Hara & Yoder, 1967; Kornprobst, 1969; Frey & Prinz, 1978; Irving, 1980; Loubet & Allègre, 1982; Griffin *et al*., 1984; Bodinier *et al*., 1987; Python & Ceuleneer, 2003). Studies of ophiolitic and orogenic peridotite massifs have also increasingly highlighted the contribution of other pyroxenite-forming processes, especially melt–peridotite interaction (Kelemen & Ghiorso, 1986; Bodinier *et al*., 1990, 2008; Kumar *et al*., 1996; Downes, 2001, 2007; Garrido *et al*., 2007; Ackerman *et al*., 2009; Lambart *et al*., 2012; Borghini *et al*., 2016).

In the Cabo Ortegal Complex, northwestern Spain, a kilometre-sized pyroxenite-rich body with single pyroxenite layers up to 3 m thick has been reported within the harzburgites of the Herbeira massif (Girardeau *et al*., 1989). Their origin remains unclear, but previous petrological and geochemical investigations have suggested a complex tectonothermal history in a subduction-related environment (Girardeau & Gil Ibarguchi, 1991; Gravestock, 1992; Santos *et al*., 2002). In this study we integrate new field observations, detailed petrography and mineralogy with whole-rock and mineral geochemistry for 30 samples. Our main aims are to (1) characterize different subtypes of pyroxenite and (2) evaluate the overprint of sub-solidus re-equilibration and fluid–rock or melt–rock interaction, in order to (3) determine a potential magmatic sequence and the composition of the melts involved and (4) speculate on the nature of their source region.

GEOLOGICAL SETTING

Relics of oceanic domains formerly separating Palaeozoic continents are preserved in the hinge of the Ibero-Armorican Arc in northwestern Iberia (Arenas *et al*., 1986; Martínez Catalán *et al*., 2009; Díez Fernández *et al*., 2013). They are regarded as remnants of a rootless suture located in the hanging wall of a thrust system, which represents an oroclinally folded section of the Variscan belt. Five allochthonous complexes are exposed (Fig. 1a), and are subdivided into an upper unit representing fragments of a Cambro-Ordovician island arc (Santos Zalduegui *et al*., 1996; Fuenlabrada *et al*., 2010), an ophiolitic unit preserving remnants of oceanic domains of various Palaeozoic ages (Pin *et al*., 2006; Arenas *et al*., 2007, 2014*b*;

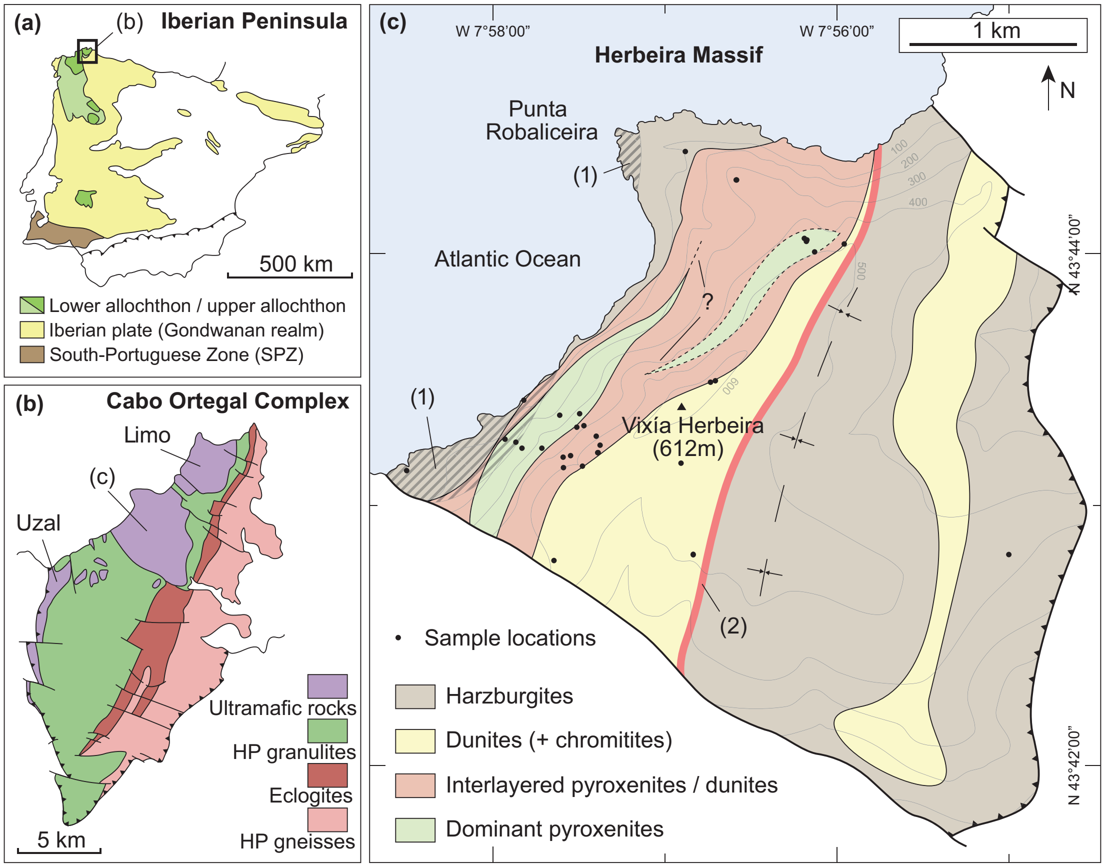

Fig. 1. (a) Location of the Cabo Ortegal Complex in the Variscan belt of the Iberian Peninsula, after Puelles *et al.* (2012); (b) details of the high-pressure units of the upper allochthon, after Llana-Fúnez *et al.* (2004); (c) geological map of the Herbeira massif showing (1) sheath-fold deformation cross-cutting lithological contacts on the western boundary of the massif and (2) the serpentinized and/or amphibolitized area surrounding the gradual contact between dunites and harzburgites, along the axial surface of the synform. Fault lines correspond either to strike-slip faults (continuous lines) or contacts with granulites (thrust patterns). Outline of the eastern dunite body is modified after Puelles *et al.* (2012).

Sánchez Martínez *et al.*, 2007), and a basal unit representing the distal Gondwanan margin (Santos Zalduegui *et al.*, 1995; Martínez Catalán *et al.*, 1996). A serpentinized mélange overlies a schistose domain and the Iberian autochthon at the base of the pile (Arenas *et al.*, 2009).

Parts of the upper allochthon and of the ophiolitic unit are exposed in the Cabo Ortegal Complex (Vogel, 1967; Marcos *et al.*, 1984; Sánchez Martínez *et al.*, 2011; Arenas *et al.*, 2014*b*). In the high-pressure–high-temperature (HP–HT) member of the former, ultramafic rocks are exposed in the Herbeira, Limo and Uzal massifs (Fig. 1b), mainly as strongly serpentinized, amphibole-bearing harzburgites with minor amounts of dunite and layered pyroxenite, representing ~10% by volume (Fig. 2c). They are associated with mafic rocks represented by mid-ocean ridge basalt (MORB)-derived eclogites (Bernard-Griffiths *et al.*, 1985; Peucat *et al.*, 1990), potentially affected by continental contamination (Gravestock, 1992), volcanic-arc granulites (Fernandez, 1994; Galán & Marcos, 1997), HP gneisses (Gil Ibarguchi *et al.*, 1990; Albert *et al.*, 2012) and their retrogressed products. Rb–Sr and Sm–Nd ages of between 487 and 506 Ma have been reported for pyroxenites and peridotites (Van Calsteren *et al.*, 1979; Santos *et al.*, 2002); this result is consistent with U–Pb zircon ages of 520–480 Ma reported for the protoliths of the eclogites and granulites (Peucat *et al.*, 1990; Ordóñez Casado *et al.*, 2001). However, the significance of these ages is debated, as they might record an ancient metamorphic event (Fernández-Suárez *et al.*, 2002, 2007). Nonetheless, a more recent magmatic event is clearly suggested by *c.* 390 Ma Rb–Sr ages (Van Calsteren *et al.*, 1979), confirmed by U–Pb (Peucat *et al.*, 1990; Santos Zalduegui *et al.*, 1996) and Sm–Nd ages (Santos *et al.*, 2002) of garnet pyroxenite dykes. This event is coeval with the metamorphic event documented at 382–395 Ma in the adjacent HP–HT units by U/Pb ages (Santos Zalduegui *et al.*, 1996; Ordóñez Casado *et al.*, 2001) and confirmed by $^{40}Ar/^{39}Ar$ ages (Gómez Barreiro *et al.*, 2007). A minimum age for the ultramafic rocks is given by a granitic pegmatite that intruded the Uzal peridotite at 382–395 Ma (Rb–Sr ages; Santos Zalduegui *et al.*, 1996).

This study is focused on the Herbeira massif, which consists of a dominantly harzburgitic plateau to the

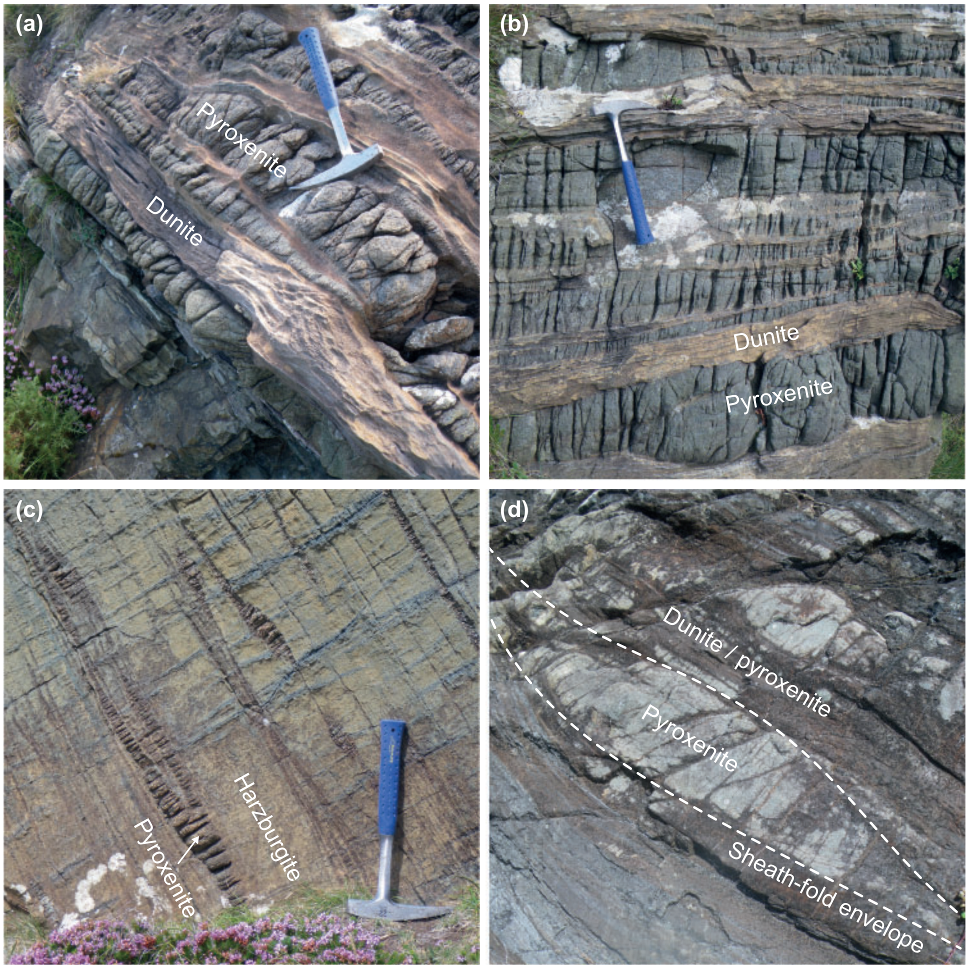

Fig. 2. Field photographs of (a) type 1 and (b) type 2 pyroxenites, typically cropping out associated with dunites along the Herbeira cliffs; (c) thinly layered pyroxenites in harzburgites; (d) the characteristic sigmoid shape of a sheath fold affecting type 3 pyroxenites and dunites at the base of the cliffs. The field of view in (d) is ~2 m.

east, similar to outcrops in the Limo and Uzal massifs, but also exposes abundant spinel- and locally garnet-bearing pyroxenites (Girardeau *et al.*, 1989) in the western cliffs. These pyroxenites are generally interlayered with dunites and are particularly abundant (~80–90%) in an ~3 km long, 300 m thick pyroxenite-rich area (Girardeau & Gil Ibarguchi, 1991), potentially consisting of one or several lenticular bodies (Fig. 1c). Dunites also occur sporadically as diffuse pods within harzburgites and in an ~600 m thick dunite-dominated zone along the edge of the cliffs, associated with chromitites (Monterrubio Pérez, 1991; Moreno *et al.*, 1999, 2001). Near the top of the pyroxenite-rich area, a 3–5 m thick, garnet-rich mafic layer with variable zoisite and rutile contents was described by Girardeau & Gil Ibarguchi (1991). Moreno (1999) and Moreno *et al.* (2001) suggested that the Trans-Herbeira Fault (THF) defines the boundary between the eastern and the western parts of the Herbeira massif (Fig. 1c). However, our field observations indicate that this contact is primarily igneous rather than tectonic. Although the area surrounding the contact is strongly serpentinized and/or amphibolitized, it is only locally brecciated and fractured and the amount of bastitized opx increases gradually over a few hundred metres between harzburgites and dunites (Fig. 1c).

The ultramafic massifs of the Cabo Ortegal Complex have experienced a high-temperature shear deformation event (Ábalos *et al.*, 2003) at >1000°C (Ben Jamaa, 1988; Girardeau *et al.*, 1990; Girardeau & Gil Ibarguchi, 1991), resulting in isoclinal folding and boudinage, and in a tectonic foliation parallel to the compositional layering (Fig. 2a–c). This is overprinted by extremely elongated sheath folds and, near tectonic contacts, by a mylonitic foliation developed during the thrusting of the ultramafic massifs over underlying granulites under high-pressure conditions (Ábalos *et al.*, 2003). Such sheath folds were described as large-scale stacks in the northwestern cliffs of the Limo massif (Puelles *et al.*, 2012); our observations indicate that they affect both peridotites and pyroxenites in the Herbeira massif (Figs 1c and 2d), which suggests that the Herbeira massif probably underwent the related deformation episode as an entity. This episode was accompanied near the peak of metamorphism (16·5–18 kbar and 800°C; Girardeau & Gil Ibarguchi, 1991) by the injection of melts that crystallized as pyroxenite and garnet-rich dykes and commonly cross-cut the compositional layering at a high angle (>30°) and are deformed only in mylonitic bands near the contacts with underlying granulites. Other pyroxenite dykes are also observed in Riedel fractures (Girardeau & Gil Ibarguchi, 1991); they are undeformed and particularly abundant in the dunite-dominated area above the pyroxenite-rich body (Fig. 1c).

Rapid exhumation (i.e. ~1 GPa in <10 Myr) occurred between 397 and 390 Ma (U/Pb ages; Fernández-Suárez *et al.*, 2002) but there is little record of the associated deformation episodes in the ultramafic massifs apart from the isoclinal folding of the Uzal peridotites and granulites. This led to the underplating and straining of the HP–HT units to form a unique assemblage at *c.* 375 Ma (Ábalos *et al.*, 2003; Albert *et al.*, 2012). Retrograde metamorphism under amphibolite- and greenschist-facies conditions accompanied the development of recumbent folds and basal thrusts (Díez Fernández *et al.*, 2011; Albert *et al.*, 2012), not recorded in the ultramafic massifs, and upright folds without axial-planar foliation, similar to the one observed in the Herbeira massif (Fig. 1c). Final emplacement of the Cabo Ortegal Complex onto the Iberian autochthon occurred at *c.* 360 Ma (Dallmeyer *et al.*, 1997).

SAMPLE DESCRIPTION

Samples were mostly collected on the western side of the Herbeira massif from pyroxenite layers ranging between <5 cm and 3 m in thickness and occurring parallel to the main tectonic foliation (Supplementary Data Electronic Appendix 1; supplementary data are available for downloading at http://www.petrology.oxfordjournals.org). These pyroxenites alternate mostly with dunites (Figs 1c and 2a, b) and locally with harzburgites (Girardeau & Gil Ibarguchi, 1991), and four types have been distinguished on the basis on their field occurrence. Type 1 pyroxenites contain dunite lenses (Fig. 2a) with sharp or slightly diffuse contacts; they are websterites, olivine websterites, olivine clinopyroxenites and clinopyroxenites. Type 2 are massive websterites (Fig. 2b). Type 3 are foliated websterites and clinopyroxenites (Fig. 2c), commonly strongly amphibolitized, which probably represent deformed (and amphibolitized) type 1 pyroxenites. Subordinate opx-rich (olivine) websterites are also present (type 4 pyroxenites). In addition to the main types of pyroxenites, millimetre- to decimetre-thick pyroxenite layers with sharp contacts with their host harzburgites (Fig. 2c) occur in lesser amounts throughout the Herbeira massif and in the Limo and Uzal massifs. These thin pyroxenite layers were not included in this study because they are commonly highly serpentinized. A comprehensive petrological description of the Cabo Ortegal peridotites, which are not included in this study, has been provided by Girardeau *et al.* (1990) and Santos *et al.* (2002).

All types of pyroxenite have medium- to coarse-grained granoblastic to porphyroclastic textures (Fig. 3), although coarse (possibly igneous) textures are locally preserved, particularly in type 1 pyroxenites (Fig. 4f). Disequilibrium textures may also be observed where unstrained amphibole has grown after cpx and spinel at the expense of matrix minerals, and on the rims and cleavages of porphyroclasts or granoblasts. This results in sutured grain boundaries and composite porphyroclastic textures (i.e. bimodal grain-size distributions), where well-equilibrated granoblastic textures may coexist within strongly non-equilibrated and sutured pyroxenes (Fig. 3c and e). This amphibole replacement, overprinting the range of igneous and (variously recrystallized) deformation-related textures, is mainly post-kinematic, as suggested for instance by undeformed sub-idiomorphic crystals nucleated within kink bands of opx porphyroclasts. The only exceptions correspond to samples collected near tectonic contacts, where amphibole may exhibit undulose extinction.

Clinopyroxene is usually well preserved and apparently unstrained (i.e. no kink bands or undulose extinction; few sub-grain boundaries) suggesting deformation at relatively high temperature and/or prolonged static recovery (Fig. 3a and b). It occurs as 1–5 mm porphyroclasts (cpx I) that locally exhibit complex boundaries and that contain exsolved needles of spinel along cleavages (Fig. 4b), as detailed below. Cpx also commonly occurs as 50–800 μm exsolution-free neoblasts (cpx II) in a matrix of porphyroclastic texture (Fig. 3b and c) or in a granoblastic texture (Fig. 3d and e). Orthopyroxene is commonly represented by variably serpentinized 2–5 mm porphyroclasts (Fig. 3c and f), which locally exhibit undulose extinction, sub-grain boundaries, kink bands and complex grain boundaries after dynamic recrystallization (opx I). These features are particularly outlined by deformed exsolved needles of spinel and cpx (Figs 3f and 4a). Opx can also be found as 50–400 μm fresh neoblasts (opx II) in much lower proportions (Fig. 3c). In opx-rich websterites, opx porphyroclasts may be fractured and these fractures filled with unexsolved opx neoblasts, and less commonly with sub-idiomorphic amphibole. Serpentinization after opx is associated with the development of late fractures and is unrelated to the type of pyroxenite. Olivine occurs mostly in type 1 pyroxenites as interstitial and elongated fine-grained trails associated with spinel (Figs 3a and 4f), or as inclusions in pyroxenes, where it provides preferential sites for the formation of sub-grain boundaries in the case of opx. Serpentine after olivine is abundant only in some type 1 pyroxenites and thinly layered type 2 and 3 pyroxenites (<5 cm) but is otherwise limited. Antigorite may also occur as late clusters of platy crystals, preferentially grown on spinel and after, or in textural equilibrium with, amphibole. Garnet is present within distinct textural assemblages of type 2 pyroxenites. It occurs either in apparent equilibrium with pyroxenes, as coronas around spinel (Figs 4c and 5a) or associated with symplectites of spinel and pyroxenes

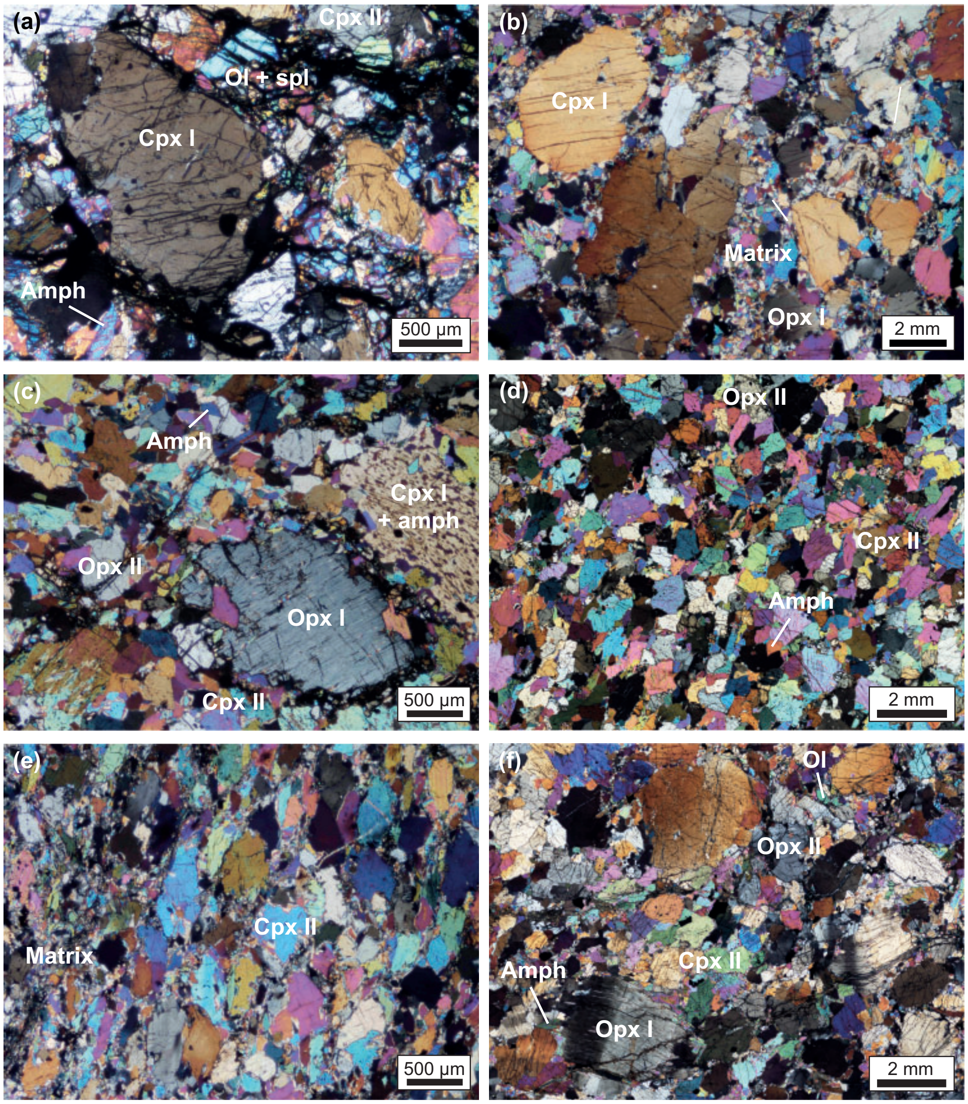

Fig. 3. Photomicrographs in cross-polarized light illustrating the textures and mineral assemblages observed in the Cabo Ortegal pyroxenites: (a) trails of interstitial olivine in a porphyroclastic olivine websterite (type 1); (b) porphyroclastic websterite (type 2); (c) abundant amphibole replacement along cleavages in a cpx porphyroclast and in the matrix of a clinopyroxenite (type 3); (d) granoblastic websterite (type 2); (e) abundant amphibole replacement along grain boundaries in a granoblastic websterite (type 3); (f) kinked opx porphyroclasts in an opx-rich websterite (type 4). (See text for further details.)

(± amphibole and ilmenite) after garnet breakdown (Fig. 4d).

Amphibole occurs in all samples, either replacing cpx (± spinel) and preferentially grown among matrix minerals, as described above (Fig. 3c), or as xenomorphic medium-sized grains in apparent textural equilibrium with cpx (Fig. 3d). In type 1 pyroxenites, amphibole and olivine are commonly associated as interstitial minerals (Figs 3a and 4e, f). Grain boundaries between these minerals are relatively fresh and the very limited serpentinization of olivine rims suggests hydration under conditions exceeding the stability field of serpentine. Where occurring within pyroxenes, and even at moderate degrees of amphibolitization, subidiomorphic to idiomorphic fine-grained crystals are observed (Fig. 5b), and are preferentially located along cleavages of cpx (Fig. 3c) and replacing cpx exsolution lamellae in opx. This elongated lamellar amphibole may be associated with spinel (Fig. 4b). At the most advanced stage of amphibolitization (mainly in type 3 pyroxenites), amphibole is observed as large, subidiomorphic to idiomorphic crystals, and in places is highly elongated. Examples of complete amphibolitization may be observed as rare hornblendites in which amphibole exhibits a poikilitic texture enclosing other amphiboles, relics of pyroxene and trails of vermicular spinel.

Spinel is mainly found as rounded or idiomorphic 20–60 µm crystals included in silicates, preferentially cpx I, opx I and amphibole (Figs 4e, f and 5b). It also occurs as

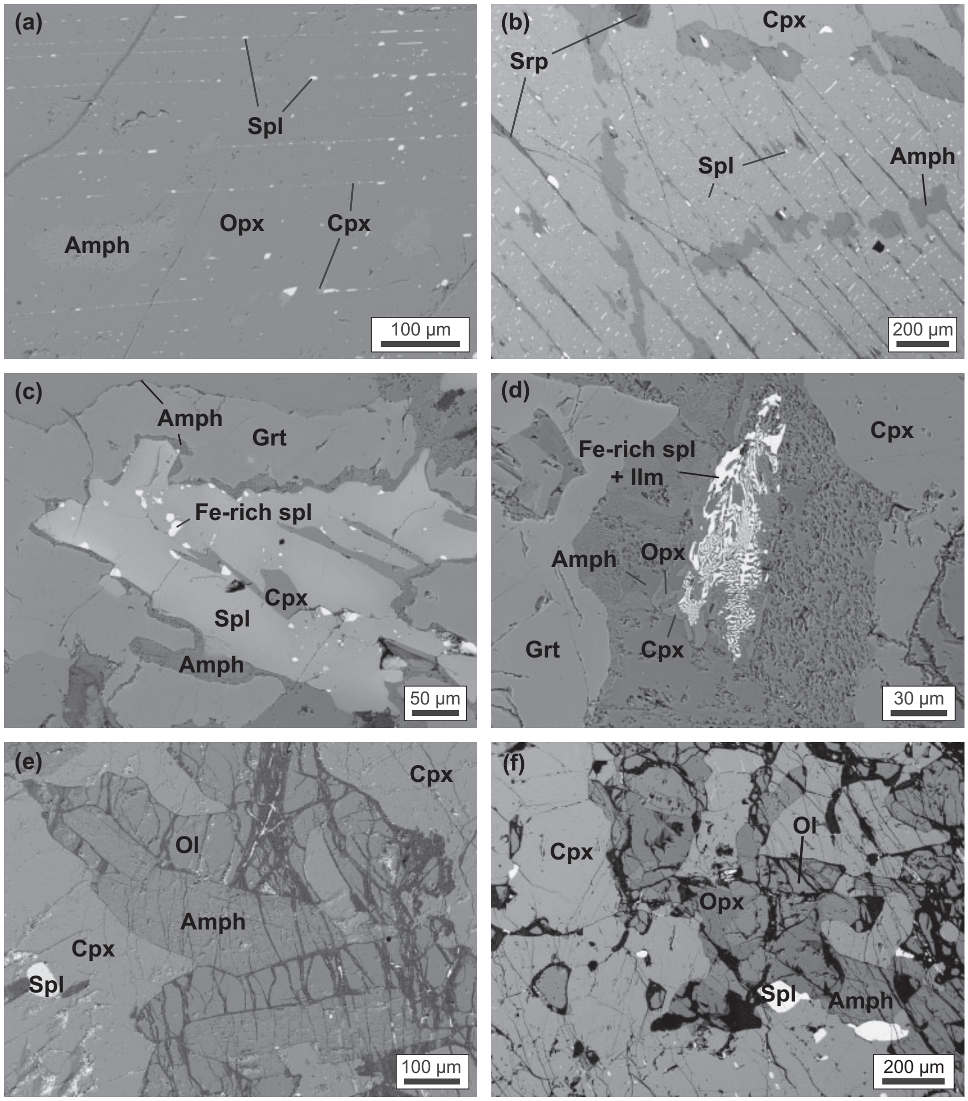

Fig. 4. Back-scattered electron images illustrating petrographic features of the Cabo Ortegal pyroxenites: (a, b) exsolved spinel along cleavages of opx and cpx, and amphibole replacement; (c) garnet corona around spinel with thin amphibole rim; (d) different symplectitic assemblages of cpx, opx, amphibole, Fe-rich spinel and ilmenite after garnet; (e) sub-idiomorphic amphibole grown into olivine trails; (f) typical textural relationships, potentially igneous, between pyroxenes and olivine with interstitial amphibole in olivine websterite (type 1).

elongated interstitial grains, locally associated with sulfides and less commonly as large idiomorphic crystals. In addition, spinel is observed as exsolved needles (± amphibole) along cleavages of pyroxene porphyroclasts (Fig. 4a and b). They may cover entire porphyroclasts or be restricted to their inner part, delineating regular- or curved-shaped cores that contrast with the sutured boundaries of the host grains (Fig. 5b). This observation may indicate pre-cooling diffusion and/or overgrowth of formerly well-equilibrated pyroxenes.

Accessory minerals include large interstitial grains of base-metal sulfides (pyrrhotite, pentlandite and chalcopyrite) with various products of serpentinization and supergene alteration. Chalcopyrite-rich grains are commonly associated with amphibole. Sulfides are also locally associated with spinel and very rarely occur as rounded homogeneous inclusions in silicates. Rare platinum-group minerals have been found, mainly Pt–Pd-rich tellurides, bismuthides and arsenides, hosted as needles in sulfides.

ANALYTICAL METHODS

Whole-rock major element compositions were determined in two batches by ALS minerals, Seville, Spain and by the Service d'Analyse des Roches et des Minéraux of the Centre de Recherches Pétrographiques et Géochimiques (CRPG), Nancy, France, using inductively coupled plasma atomic emission spectroscopy (ICP-AES); loss on ignition (LOI) was determined at

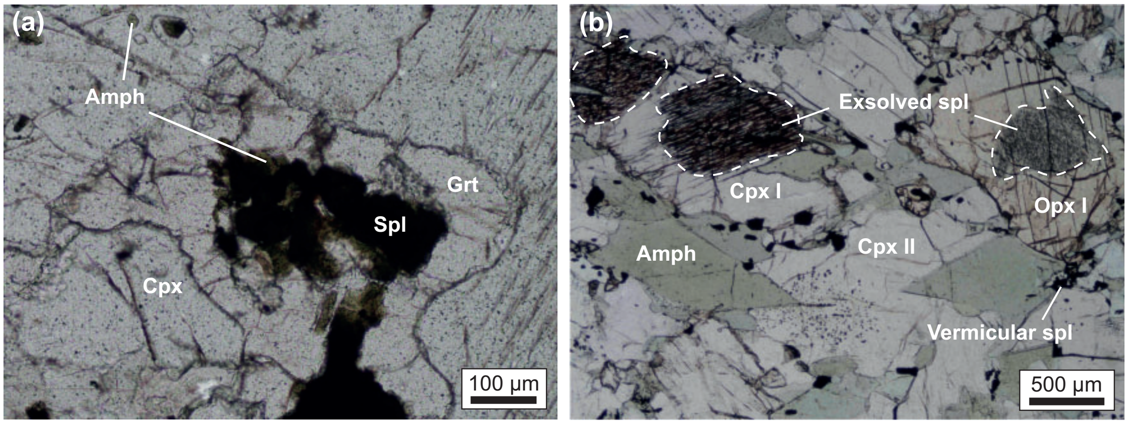

Fig. 5. Photomicrographs in plane-polarized light illustrating sub-solidus assemblages observed in the Cabo Ortegal pyroxenites: (a) undeformed garnet with thin layer of amphibole around spinel in a type 2 websterite (also shown in Fig. 4c); (b) sub-idiomorphic to idiomorphic amphibole after cpx and spinel, and exsolved spinel restricted to grain cores in a type 2 websterite; the vermicular spinel associated with amphibole should be noted.

1000°C for each sample. Whole-rock trace element compositions also were analysed in two batches by solution inductively coupled plasma mass spectrometry (ICP-MS) on a Agilent 7500ce system and on a high-resolution (HR) ICP-MS Element XR system at Géosciences Environnement Toulouse (GET), Toulouse, France. Homogeneous sections of sample were cleared of patina, crushed and powdered in an agate mill. Powders were digested following method C of Yokoyama *et al.* (1999) and spiked with a Tm solution following Barrat *et al.* (2012). USGS standard BHVO-2 was used to check instrumental drift and analyses of IWG-GIT standard PM-S were included as unknown samples for quality control (Supplementary Data Electronic Appendix 2a). Whole-rock compositions were also reconstructed from mineral data and point-counted modes. Modal proportions by volume were estimated using ~2000 points at 0·5–0·8 mm steps on a thin section, and were then converted to mass proportions by calculating mineral densities from the molar ratios of their end-members. Mineral proportions were confirmed using a least-squares method modified from MINSQ (Herrmann & Berry, 2002), using measured whole-rock and averaged mineral major element compositions.

Electron microprobe analysis (EMPA) and laser ablation (LA)-ICP-MS were performed at the Geochemical Analysis Unit (GAU) of CCFS–GEMOC at Macquarie University, Australia. For that purpose, polished thin (~30 μm) and thick (~100 μm) sections were prepared and examined under an optical microscope. Backscattered electron (BSE) images were taken on a Zeiss EVO MA15 scanning electron microscope (SEM) using an accelerating voltage of 15 kV and a beam current of 20 nA. Mineral major element compositions were determined by EMPA using a CAMECA SX100 instrument fitted with five wavelength-dispersive spectrometers, at an accelerating voltage of 15 kV, a beam current of 3 nA and a beam size of 1–2 mm, counting 10 s on peaks and 5 s on background on each side of the peak. Natural minerals and synthetic oxides were used as standards following the PAP correction method of Pouchou & Pichoir (1984). Relative standard deviations on standard analyses are better than 1%. Trace element compositions of cpx, opx, amphibole, olivine and garnet were determined by LA-ICP-MS using a Photon Machine Excite 193 nm ArF Excimer laser attached to an Agilent 7700 ICP-MS system. Beam size ranged between 30 and 60 μm at a pulse rate of 5 Hz and an energy density of 3·0–4·4 J·cm^{-2}. Each analytical sequence started with 60 s of background, 120 s during ablation and 30 s of washout. NIST SRM610 was used as an external standard and ^{43}Ca was taken as the internal standard for cpx, garnet and amphibole whereas ^{29}Si was used for opx and olivine. The data reduction software GLITTER 4.4 (Griffin *et al.*, 2008) was used to calibrate the fractionation induced by ablation, transportation and excitation processes and the matrix effect. Analyses of USGS standards BCR-2G and BHVO-2 were included as unknown samples in each batch for quality control (Supplementary Data Electronic Appendix 2b).

RESULTS

Whole-rock compositions and mineral modes

Point-counted modes are consistent with mineral modes calculated using the least-squares method (Table 1), although the latter tends to give slightly lower cpx/opx and higher olivine proportion than the former. Point counting yields 32–80 wt % cpx (10–23 wt % in opx-rich, type 4 websterites) and 0–45 wt % opx (65–79 wt % in type 4 websterites). Pyroxenites with dunite

Table 1: Point-counted and calculated modal compositions (wt %) of Cabo Ortegal pyroxenites and peridotites

Mineral	Cpx	Opx	Ol	Amph	Spl	Grt	Sulf	Srp	Atg	RSSQ	Total
Pyroxenites with dunite lenses (*type 1*)											
CO-012-A	57 (57)	27 (24)	0 (11)	6 (2)	7 (6)		< 0.5	1	1	0.04	99
CO-013-A	62 (54)	6 (7)	0 (18)	9 (13)	5 (4)		1	16	2	0.01	95
CO-013-C	63 (46)	14 (15)	0.3 (14)	8 (22)	4 (0)		0.5	8	2	0.03	97
CO-066-B2	55	18	6	11	4		1	4			
CO-091-A	72 (67)	1 (10)	3 (10)	11 (2)	5 (5)		< 0.5	7		0.30	95
CO-094-B	80 (56)	0.6 (9)	5 (20)	8 (8)	4 (4)			2	1	0.09	98
CO-095-A	74 (80)	12 (14)	2 (1)	7 (0)	3 (3)		1	2		1.08	97
CO-096-A	63	10	7	14	3			2			
CO-096-B	70 (59)	15 (12)	2 (11)	8 (14)	4 (4)		< 0.5	1		0.54	99
CO-100	42	9	15	13	6		< 0.5	15			
Massive pyroxenites (*type 2*)											
CO-006-A	45 (51)	25 (30)	0 (1)	26 (13)	3 (3)		< 0.5	< 0.5		0.41	98
CO-006-B	46 (54)	26 (21)	0 (2)	16 (17)	3 (3)	7 (1)	< 0.5	< 0.5		0.00	98
CO-007	56 (53)	29 (24)		11 (19)	3 (3)		< 0.5	2		0.72	100
CO-019-A	34 (57)	39 (22)	0 (1)	20 (17)	3 (3)		1	4		0.16	99
CO-024	34 (42)	36 (22)	0 (0.1)	18 (33)	2 (2)	6 (1)	1	2	< 0.5	0.16	100
CO-025	60 (60)	13 (25)	0 (1)	19 (7)	5 (5)		1	2		0.04	98
CO-045-A	43 (54)	45 (31)	0 (2)	8 (7)	2 (2)		1	< 0.5		0.07	97
CO-046-A	66 (70)	12 (15)	0 (0.1)	11 (5)	8 (6)		1	2		0.07	96
CO-046-C	57	5		10	5		1	22			
CO-065	55	8	5	5				28	< 0.5		
Foliated pyroxenites (*type 3*)											
CO-004-A	57 (58)	0 (5)		43 (37)				< 0.5		0.07	100
CO-009	69 (77)	10 (3)	0 (8)	11 (5)	6 (5)		1	1	2	0.03	99
CO-010-A	51	3	3	13	4		2	25			
CO-010-B	47 (56)	10 (10)	0 (25)	8 (4)	5 (1)		< 0.5	30		0.01	95
CO-044-C	31 (32)	38 (38)		24 (28)	5 (2)		1	2		0.88	99
CO-067	64 (47)	6 (20)		26 (29)	3 (3)		< 0.5	1		0.46	99
CO-098	55 (48)	17 (15)	0 (2)	21 (31)	2 (2)		< 0.5	5		0.06	98
CO-101	46 (33)	12 (13)		38 (50)	3 (3)		1	< 0.5		0.66	98
Opx-rich websterites (*type 4*)											
CO-002-A	19	79	1	5	1		1	4			
CO-048	23 (5)	65 (80)	4 (6)	< 0.5	0.5 (< 0.5)		< 0.5	2		3.64	95

Mineral modes were estimated from point-counted modes and converted to wt % from mineral densities using $d_{cpx} = 3{\cdot}1$, $d_{opx} = 3{\cdot}3$, $d_{ol} = 3{\cdot}2$, $d_{amph} = 3{\cdot}2$, $d_{spl} = 4{\cdot}6$, $d_{grt} = 3{\cdot}8$, d_{srp} and $d_{atg} = 2{\cdot}6$ (see text for further details). Least-squares fitted modes (results in parentheses) were calculated from whole-rock and mineral major element compositions using MINSQ (Herrmann & Berry, 2002). The solution method of Solver, in Microsoft Excel™ was set to the GRG Nonlinear engine for most samples, and to the Evolutionary engine for samples where satisfactory residual sum of squares (RSSQ) values (i.e. < 1) could not be reached. A constrained minimization of 0–100 wt % was used for all minerals apart from spinel, which was constrained to a maximum set to the counted value when available, or arbitrarily to 5 wt %. Total values are related to the least-squares method calculation. All samples are from Herbeira massif apart from: *from Limo massif; †from Uzal massif.

lenses (type 1) and foliated (type 3) pyroxenites have higher cpx/opx than massive (type 2) pyroxenites (Fig. 6). Olivine is mostly restricted to type 1 pyroxenites with the exception of thin pyroxenite layers (<5 cm) where its presence may reflect mechanical mixing with the host peridotites (Fig. 6). The modal proportion of amphibole ranges from 6–14 wt % in type 1 and 4–26 wt % in type 2 pyroxenites to 8–42 wt % in type 3 pyroxenites. Spinel represents 2–8 wt % (<1 wt % in opx-rich websterites), garnet 6–7 wt % when present, and sulfides never exceed 1 wt %.

Measured whole-rock major and trace element compositions of the Cabo Ortegal pyroxenites are reported in Table 2. The major element compositions of type 1, 2 and 3 pyroxenites are intermediate between those of the Cabo Ortegal peridotites and of the garnet-rich mafic layer located near the top of the pyroxenite-rich body, except for much higher SiO_2 and lower Al_2O_3 (Fig. 7). The content of most major oxides is directly related to modal variations. Opx-rich (type 4) websterites accordingly have higher MgO and SiO_2, and lower CaO, Al_2O_3, and TiO_2 than other types of pyroxenites, which is similar to the composition of the Cabo Ortegal peridotites apart from higher SiO_2. These data overlap the range of compositions previously reported for Cabo Ortegal pyroxenites (Gravestock, 1992), with type 1 pyroxenites having slightly lower SiO_2 and higher MgO. Compared with pyroxenite xenoliths and pyroxenites from orogenic and ophiolitic massifs worldwide (Fig. 7) they are very rich in CaO (11·2–20·1 wt %), SiO_2 (47·0–52·7 wt %) and Cr_2O_3 (0·23–1·93 wt %) and have low TiO_2 (<0·20 wt %) and Al_2O_3 (1·29–6·59 wt %), resulting in particularly high CaO/Al_2O_3 (2·2–11·3) and Cr# [molar Cr/(Al + Cr); 0·04–0·37]. They also have relatively high Mg# [molar $Mg/(Mg + Fe^{tot})$] ranging between 0·83 and 0·90, with type 1 pyroxenites restricted to high values

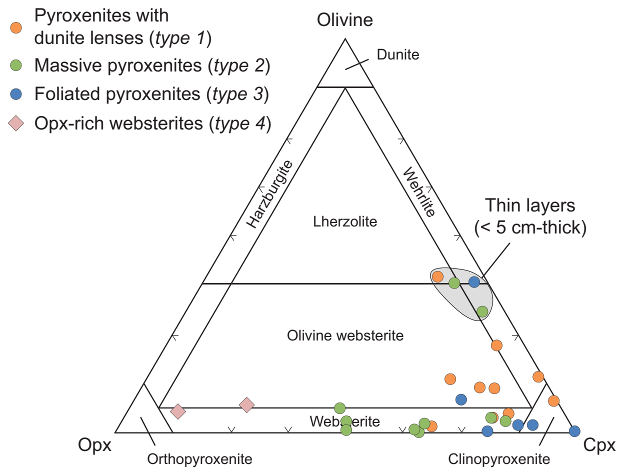

Fig. 6. Point-counted modal compositions of Cabo Ortegal pyroxenites recalculated into olivine (including serpentine), clinopyroxene (Cpx) and orthopyroxene (Opx). Thin layers (<5 cm thick) are indicated—their modal composition may be strongly affected by mechanical mixing with the host peridotites.

(0·87–0·89) and overlapping the range of Mg# in Cabo Ortegal dunites and in certain harzburgites. Alkalis are generally very low in all types of pyroxenite, including type 4 pyroxenites, although K_2O reaches 0·27 wt % in amphibole-rich, type 3 pyroxenites. The abundance of transition metals is related to modal variations. Sc, V and Zn are the highest in type 2 pyroxenites and lowest in type 1 and 4 pyroxenites, whereas type 3 pyroxenites overlap the whole range of compositions. Co, Ni and particularly Cr show less clear variation although they are slightly more abundant in type 1 and 4 pyroxenites. Cu is unrelated to the types of pyroxenite, suggesting that the proportion of Cu-bearing sulfides is also not related to the modal proportion of silicates.

Whole-rock trace element compositions reconstructed from mineral modes and major element compositions are consistent with the measured whole-rock compositions for most samples (Fig. 8). They consistently show that the different types of pyroxenite have very distinctive rare earth element (REE) patterns (Fig. 9). Most type 1 pyroxenites have characteristic spoon-shaped patterns with flat middle REE (MREE) to heavy REE (HREE) [$(Gd/Lu)_{CN} = 0{\cdot}73$–$1{\cdot}3$; CN indicates chondrite-normalized], whereas type 3 pyroxenites have enriched and strongly fractionated light REE (LREE) and MREE [$(La/Gd)_{CN} = 2{\cdot}6$–$8{\cdot}6$]. Exceptions with LREE-enriched patterns among type 1 pyroxenites include a sample collected near a pyroxenite dyke with dunitic rims (CO-091-A). Type 2 pyroxenites exhibit a range of REE patterns with progressive LREE enrichment [$(La/Gd)_{CN} = 1{\cdot}1$–$3{\cdot}2$], including a very LREE-depleted sample (CO-006-A) with $(La/Gd)_{CN} < 0{\cdot}01$ and a sample (CO-045-A) with a saddle-shaped REE pattern. Most LREE-enriched type 2 and 3 samples also exhibit slightly negative slopes from MREE to HREE [$(Gd/Lu)_{CN} = 1{\cdot}1$–$2{\cdot}0$]. Among all types, only the least enriched type 2 samples and two type 3 samples exhibit slightly positive Eu anomalies [$(Eu/Eu^*)_{CN} = 1{\cdot}1$–$1{\cdot}4$].

In a multi-element diagram normalized to primitive-mantle values (PM), Cabo Ortegal pyroxenites exhibit a strong enrichment of most large ion lithophile elements (LILE: Cs, Rb, Ba, K, Pb and Sr) and other fluid-mobile elements (Th, U and Li) relative to neighbouring lithophile elements (Fig. 10). This enrichment is pronounced, although variable (regardless of the type of pyroxenite) for Cs, Ba, U, Pb and Sr. Variations in Li content in type 2 and 3 pyroxenites result in zero to strongly positive anomalies relative to neighbouring REE [e.g. $(Li/Dy)_{PM} = 1{\cdot}2$–$7{\cdot}1$], whereas type 1 pyroxenites have weak anomalies [$(Li/Dy)_{PM} < 1{\cdot}4$, with one sample reaching 4·3]. The high field strength elements (HFSE: Nb, Ta, Zr, Hf and Ti) are strongly depleted, with particularly low Ta [$(Nb/Ta)_{PM} = 0{\cdot}9$–$3{\cdot}1$] and low Zr [$(Zr/Hf)_{PM} = 0{\cdot}01$–$0{\cdot}97$]. All type 3 pyroxenites exhibit negative Ti anomalies [$(Ti/Ti^*)_{PM} = 0{\cdot}36$–$0{\cdot}57$], whereas type 1 and 2 pyroxenites have either negative or slightly positive anomalies [$(Ti/Ti^*)_{PM} = 0{\cdot}36$–$1{\cdot}2$].

Mineral chemistry

Major elements

Representative major element compositions of cpx, opx, amphibole, olivine and garnet are reported in Table 3. Cpx has higher Mg# (0·91–0·94) in type 1 than in type 2 pyroxenites (0·88–0·92), which have the highest Al_2O_3 (1·12–2·79 wt %) and TiO_2 (up to 0·25 wt %), outlining a well-defined trend on Al_2O_3 (Fig. 11a) or TiO_2 vs Mg# plots. The Mg# of cpx in type 3 pyroxenites overlaps that of the first two types of pyroxenite, although a significant proportion of these values are $> 0{\cdot}92$. However, their Al_2O_3 is lower by $\sim 0{\cdot}4$ wt % (Fig. 11a); TiO_2 is also slightly lower. Other differences in cpx compositions include the lowest SiO_2 (51·7–54·5 wt %) and highest CaO (24·1–25·1 wt %) in type 1 pyroxenites. Cpx in type 4 pyroxenites mainly differs from that of type 1 by lower Al_2O_3 (0·47–1·15 wt %), resulting in higher Cr#. The compositions of cpx from type 1 and 4, and most type 3 pyroxenites overlap those of cpx from dunites for most oxides. Within single samples, cores of cpx I commonly have higher Al_2O_3 (by up to 1 wt %), and to a minor extent TiO_2, and lower SiO_2 and Mg# than rims. This difference represents on average $\sim$0·05 in Mg# and 0·3–0·4 wt % in Al_2O_3 and is not systematically observed in strongly amphibolitized samples. Cpx II may have similar composition to the rims of cpx I or lower Al_2O_3. No significant variation was observed between exsolved and exsolution-free core of cpx porphyroclasts.

The Mg# of opx increases similarly to cpx from type 1 to type 2 pyroxenites, with type 3 being intermediate between these two (Fig. 11b). A trend of increasing Al_2O_3 with decreasing Mg#, similar to that in cpx, is observed in opx, with opx in type 3 pyroxenites having slightly lower Al_2O_3. Opx from type 4 pyroxenites has a narrow range of high Mg# ($\sim$0·89) and lower Al_2O_3

Table 2: Whole-rock major (wt %) and trace element (ppm) compositions of Cabo Ortegal pyroxenites

Rock type:	Pyroxenites with dunite lenses (type 1)						
Sample:	CO-012-A*	CO-013-A*	CO-013-C*	CO-091-A†	CO-094-B†	CO-095-A†	CO-096-B†
SiO_2	51·90	47·00	48·50	48·83	48·37	52·16	49·82
TiO_2	0·08	0·08	0·08	0·08	0·09	0·06	0·08
Al_2O_3	2·55	3·14	3·39	2·16	2·69	1·78	2·94
Cr_2O_3	0·84	0·75	0·64	n.a.	n.a.	n.a.	n.a.
FeO_t	5·44	5·41	5·23	4·97	6·26	4·77	5·16
MnO	0·13	0·12	0·12	0·12	0·13	0·13	0·12
MgO	24·10	22·90	23·60	21·90	24·16	19·28	22·58
CaO	14·75	14·95	13·90	17·47	15·23	20·04	16·70
Na_2O	0·17	0·20	0·26	0·17	0·14	0·20	0·31
K_2O	0·02	0·02	0·02	0·03	0·01	0·02	0·04
P_2O_5	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.
LOI	0·95	5·45	5·21	3·40	1·93	0·93	1·58
Total	101·54	100·64	101·54	99·68	99·72	99·89	99·91
Mg#	0·89	0·88	0·89	0·89	0·87	0·88	0·89
Li	1·4	1·9	1·9	1·0	1·0	2·6	0·7
Be	n.a.	n.a.	n.a.	0·50	b.d.l.	0·40	0·16
B	5·0	8·0	8·0	12·3	13·2	13·5	10·9
Sc	159	189	211	41	38	44	29
V	135	189	202	256	252	240	181
Cr	n.a.	n.a.	n.a.	6194	3959	4833	6616
Co	197	71	71	43	55	43	45
Ni	764	1125	781	527	500	380	633
Cu	183	353	159	122	68	182	278
Zn	19·3	29·2	24·0	19·6	27·6	19·5	14·5
Ga	1·71	2·27	2·78	1·51	1·99	1·67	1·56
Rb	0·86	0·43	0·63	0·30	0·27	0·53	0·46
Sr	17·9	27·1	38·5	54·9	8·7	33·1	16·6
Y	3·17	3·87	4·07	2·41	2·60	1·47	2·08
Zr	0·81	0·94	2·39	2·02	1·75	2·46	1·50
Nb	n.a.	n.a.	n.a.	n.a.	0·05	0·06	0·10
Cs	0·09	0·14	0·15	0·03	0·04	0·10	0·03
Ba	9·2	12·1	17·5	11·2	6·3	6·2	13·5
La	0·33	0·23	0·39	1·09	0·22	1·10	0·20
Ce	0·36	0·36	1·02	2·37	0·44	2·09	0·36
Pr	0·06	0·06	0·15	0·26	0·06	0·22	0·04
Nd	0·39	0·34	0·67	1·03	0·36	0·80	0·24
Sm	0·20	0·17	0·20	0·23	0·17	0·17	0·12
Eu	0·08	0·07	0·08	0·09	0·07	0·07	0·05
Gd	0·33	0·37	0·38	0·31	0·28	0·22	0·20
Tb	0·06	0·08	0·09	0·06	0·06	0·04	0·04
Dy	0·50	0·59	0·64	0·40	0·43	0·25	0·32
Ho	0·11	0·13	0·15	0·09	0·10	0·05	0·08
Er	0·34	0·41	0·44	0·25	0·28	0·16	0·22
Tm	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Yb	0·32	0·40	0·46	0·22	0·25	0·15	0·21
Lu	0·05	0·06	0·07	0·03	0·04	0·02	0·03
Hf	0·10	0·09	0·14	0·08	0·09	0·08	0·07
Ta	n.a.	n.a.	n.a.	n.a.	0·002	0·003	0·004
Pb	0·3	0·9	1·1	1·0	0·4	1·1	0·6
Th	0·42	0·33	0·18	0·08	0·06	0·12	0·06
U	0·11	0·10	0·11	0·08	0·06	0·09	0·05

(continued)

(0·39–1·42) than other types of pyroxenite. Opx in dunites is comparable with that in type 1 and 4 pyroxenites (Santos *et al.*, 2002). However, opx in harzburgites defines a markedly different trend, reaching high Al_2O_3 (7·13 wt %) and CaO (3·19 wt %) over a narrow range of Mg# (0·86–0·92), comparable with that in dunites (0·89–0·91). Compositional variations between core and rim of porphyroclasts are significant only in terms of Al_2O_3 (up to 1·5 wt %) and SiO_2, whereas Mg# is similar within $\pm$0·008 units. This results in $\sim$0·2 wt % difference in Al_2O_3 on average and is not observed in areas where large patches of matrix minerals surround porphyroclasts, which then have relatively homogeneous core-like composition. No additional variation associated with the presence of spinel ($\pm$ amphibole) exsolution was observed, as in cpx.

Olivine in type 1 and 4 pyroxenites has lower Mg# (0·86–0·89) than olivine in dunites and harzburgites (Fig. 11d). NiO in olivine increases with Mg# from type 1 pyroxenites to dunites and harzburgites whereas olivine has higher NiO (0·27–0·48 wt %) in type 4 than type 1 pyroxenites at comparable Mg#. When plotted against

Table 2: Continued

Rock type:	Massive pyroxenites (type 2)							
Sample:	CO-006-A*	CO-006-B†	CO-007*	CO-019-A*	CO-024*	CO-025*	CO-045-A*	CO-046-A*
SiO_2	52·20	51·30	52·00	52·70	51·20	52·50	51·60	50·70
TiO_2	0·09	0·11	0·13	0·10	0·15	0·11	0·12	0·15
Al_2O_3	3·03	3·26	3·82	3·25	5·89	3·13	2·77	3·38
Cr_2O_3	0·51	n.a.	0·61	0·58	0·2	0·5	0·3	0·6
FeO_t	7·27	7·07	4·98	5·44	6·73	6·17	6·12	5·57
MnO	0·16	0·18	0·12	0·13	0·16	0·15	0·12	0·13
MgO	20·40	19·85	20·70	20·30	20·20	19·95	21·70	18·05
CaO	14·35	15·49	15·35	16·25	14·60	15·90	14·40	18·20
Na_2O	0·44	0·19	0·55	0·49	0·30	0·37	0·22	0·38
K_2O	0·03	0·02	0·03	0·04	0·02	0·04	0·02	0·02
P_2O_5	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.
LOI	0·99	0·48	1·39	1·26	0·88	1·05	1·09	0·97
Total	100·29	98·74	100·24	101·17	101·12	100·56	99·18	98·76
Mg#	0·83	0·83	0·88	0·87	0·84	0·85	0·86	0·85
Li	1·9	2·4	2·3	2·4	1·7	2·0	1·2	1·8
Be	n.a.	0·63	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
B	5·0	18·1	5·0	b.d.l.	b.d.l.	6·0	b.d.l.	5·0
Sc	281	73	318	230	246	277	211	332
V	209	438	283	256	241	244	231	261
Cr	n.a.	2707	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Co	86	74	76	57	61	58	108	56
Ni	358	217	799	531	230	353	770	267
Cu	95	174	232	197	41	118	524	208
Zn	31·6	34·2	25·2	19·8	43·0	30·5	29·0	22·6
Ga	3·24	3·52	4·24	11·60	4·58	3·88	3·25	2·54
Rb	0·50	0·61	1·05	0·84	0·78	0·65	0·63	0·55
Sr	12·6	57·5	44·9	48·8	42·0	89·7	49·3	47·3
Y	2·08	3·33	2·41	3·07	2·78	3·74	1·37	3·46
Zr	1·00	5·40	1·51	1·15	3·11	1·89	1·42	5·31
Nb	0·06	n.a.	n.a.	n.a.	0·11	0·56	n.a.	n.a.
Cs	0·05	0·07	0·15	0·27	0·16	0·12	0·09	0·10
Ba	21·9	14·0	23·5	123·0	22·0	26·3	19·0	13·3
La	0·00	0·91	1·30	0·46	0·82	2·05	0·44	0·91
Ce	0·17	2·80	1·47	0·55	1·82	7·81	0·74	1·95
Pr	0·05	0·42	0·12	0·07	0·22	1·22	0·11	0·28
Nd	0·36	2·00	0·50	0·40	1·01	4·91	0·52	1·26
Sm	0·16	0·51	0·24	0·17	0·30	0·89	0·24	0·39
Eu	0·09	0·17	0·14	0·10	0·15	0·28	0·10	0·17
Gd	0·24	0·53	0·35	0·34	0·43	0·71	0·31	0·49
Tb	0·05	0·08	0·06	0·06	0·08	0·11	0·05	0·09
Dy	0·34	0·54	0·42	0·52	0·51	0·66	0·30	0·60
Ho	0·08	0·12	0·09	0·11	0·11	0·13	0·05	0·13
Er	0·24	0·35	0·27	0·35	0·29	0·42	0·12	0·39
Tm	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Yb	0·23	0·34	0·25	0·34	0·27	0·37	0·08	0·35
Lu	0·04	0·05	0·04	0·05	0·04	0·05	0·01	0·05
Hf	0·14	0·20	0·19	0·10	0·20	0·09	0·09	0·20
Ta	0·002	n.a.	n.a.	n.a.	0·002	0·014	n.a.	n.a.
Pb	0·7	1·4	1·2	1·9	0·7	0·9	1·1	1·0
Th	0·30	0·09	0·84	0·21	0·23	0·31	0·09	0·22
U	0·06	0·14	0·27	0·19	0·13	0·24	0·22	0·20

(continued)

Cr# of spinel from corresponding samples, Mg# of olivine from pyroxenites is shifted by 0·02–0·04 Mg# units from the olivine–spinel mantle array (OSMA: Arai, 1987, 1994), whereas olivine from dunite falls within this array (Santos *et al.*, 2002).

Amphibole shows compositional differences between the different types of pyroxenites that are comparable with those of cpx. Al_2O_3 and TiO_2 increase with decreasing Mg# from type 1 to type 2 pyroxenites, with type 3 overlapping this range of compositions, and extending it to slightly lower Mg#. Accordingly, amphibole also has lower SiO_2 in type 1 pyroxenites. In addition, K_2O (0·12–0·69 wt %) and, to a lesser extent, Na_2O are systematically higher in amphibole from type 3 pyroxenites (Fig. 11c). Within single samples, there is no systematic compositional difference between the different textures of amphibole.

Trace elements

Representative trace element compositions of cpx and amphibole are reported in Tables 4 and 5 respectively.

Table 2: Continued

Rock type:	Foliated pyroxenites (type 3)							Type 4 pyrox.
Sample:	CO-004-A*	CO-009*	CO-010-B*	CO-044C*	CO-067†	CO-098†	CO-101†	CO-048*
SiO_2	52·20	51·90	47·80	52·10	52·00	49·93	48·74	53·70
TiO_2	0·20	0·05	0·04	0·07	0·08	0·15	0·12	0·02
Al_2O_3	4·39	2·30	1·29	3·15	2·92	4·39	6·59	0·89
Cr_2O_3	0·40	0·80	0·71	0·42	n.a.	n.a.	n.a.	0·56
FeO_t	5·36	4·05	5·57	6·43	5·99	5·18	6·62	8·98
MnO	0·15	0·12	0·12	0·15	0·15	0·11	0·16	0·17
MgO	17·40	19·60	25·60	23·00	20·66	20·26	19·18	31·70
CaO	18·60	20·10	14·25	11·15	14·83	15·99	14·42	2·10
Na_2O	0·73	0·33	0·08	0·47	0·58	0·59	0·78	0·09
K_2O	0·19	0·04	b.d.l.	0·09	0·13	0·16	0·27	0·03
P_2O_5	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.
LOI	1·59	1·48	5·39	1·33	1·92	2·82	1·91	1·03
Total	101·87	101·24	101·48	99·12	99·93	100·16	99·53	98·28
Mg#	0·85	0·90	0·89	0·86	0·86	0·88	0·84	0·89
Li	3·9	1·4	1·3	1·8	3·5	5·8	2·0	5·4
Be	n.a.	n.a.	n.a.	n.a.	2·52	0·88	2·36	n.a.
B	5·0	6·0	7·0	b.d.l.	12·9	12·7	13·2	5·0
Sc	395	292	171	186	37	36	60	10
V	246	191	155	183	255	224	305	45
Cr	n.a.	n.a.	n.a.	n.a.	3475	3835	2446	n.a.
Co	52	44	71	80	42	51	46	81
Ni	258	428	1022	417	373	352	257	843
Cu	74	109	202	92	170	85	159	109
Zn	29·0	10·6	23·1	31·4	21·9	17·1	16·9	38·1
Ga	8·04	3·54	1·20	5·83	2·25	3·20	4·10	3·49
Rb	1·58	0·57	0·10	1·61	1·11	2·54	2·83	1·41
Sr	424·7	97·4	16·1	108·0	146·0	103·6	138·6	79·6
Y	6·31	2·55	1·68	1·96	1·61	1·90	4·58	0·31
Zr	5·47	2·08	0·01	5·14	4·72	12·96	15·39	0·39
Nb	0·80	n.a.	n.a.	n.a.	n.a.	n.a.	1·19	0·08
Cs	0·03	0·05	0·02	0·14	0·07	0·26	0·16	0·99
Ba	68·6	28·2	7·1	54·5	28·1	54·4	84·5	33·9
La	7·27	3·40	0·55	1·64	2·07	2·44	1·95	0·81
Ce	14·09	5·58	0·34	3·66	4·42	5·15	4·48	1·91
Pr	1·69	0·52	0·02	0·48	0·55	0·55	0·61	0·23
Nd	6·68	1·52	0·11	2·03	2·18	2·03	2·73	0·78
Sm	1·22	0·26	0·08	0·44	0·38	0·38	0·62	0·10
Eu	0·40	0·10	0·04	0·12	0·13	0·16	0·23	0·03
Gd	1·17	0·33	0·16	0·37	0·34	0·40	0·63	0·08
Tb	0·17	0·07	0·04	0·06	0·05	0·06	0·10	0·01
Dy	1·12	0·44	0·26	0·35	0·26	0·34	0·69	0·05
Ho	0·23	0·09	0·06	0·07	0·05	0·07	0·16	0·01
Er	0·63	0·27	0·19	0·21	0·16	0·20	0·47	0·02
Tm	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Yb	0·54	0·25	0·17	0·21	0·15	0·17	0·49	0·05
Lu	0·08	0·04	0·02	0·03	0·02	0·02	0·07	0·01
Hf	0·30	0·12	0·06	0·21	0·15	0·36	0·75	0·07
Ta	0·021	n.a.	n.a.	n.a.	n.a.	n.a.	0·081	0·003
Pb	2·5	1·6	0·7	1·3	2·1	1·6	3·2	0·4
Th	0·51	0·52	0·40	0·28	0·21	0·30	0·23	0·07
U	0·28	0·20	0·11	0·32	0·16	0·14	0·36	0·05

*Major element compositions were determined by solution ICP-AES after $LiBO_2$ fusion at ALS Minerals, Seville, Spain; trace element compositions (except Li and B) were determined by solution ICP-MS at GET, Toulouse, France; Li and B were analysed at SARM, CRPG, Nancy, France by flame atomic absorption spectroscopy (AAS) after acid digestion and by UV–Vis after Na_2Co_3 fusion, respectively.
†Major element compositions (except Cr_2O_3) were determined after $LiBO_2$ fusion by solution ICP-AES, and Cr by ICP-MS, at SARM, CRPG, Nancy, France; trace element compositions (including Li, B and Be) were determined by solution ICP-MS at GET, Toulouse, France.
FeO_t, total Fe as FeO; LOI, loss on ignition; b.d.l., below detection limit; n.a., not analyzed; Mg#, molar ratio of $Mg/(Mg + Fe^{tot})$.

Trace element compositions of olivine and opx are not discussed here as they host a negligible proportion of the whole-rock budget because of their very low concentrations (commonly below or near detection limit) and/or modal proportions. They have depleted REE patterns and very low contents of most lithophile trace elements, and only certain LILE and other fluid-mobile elements (e.g. Cs, U, Th and Li) may reach near-

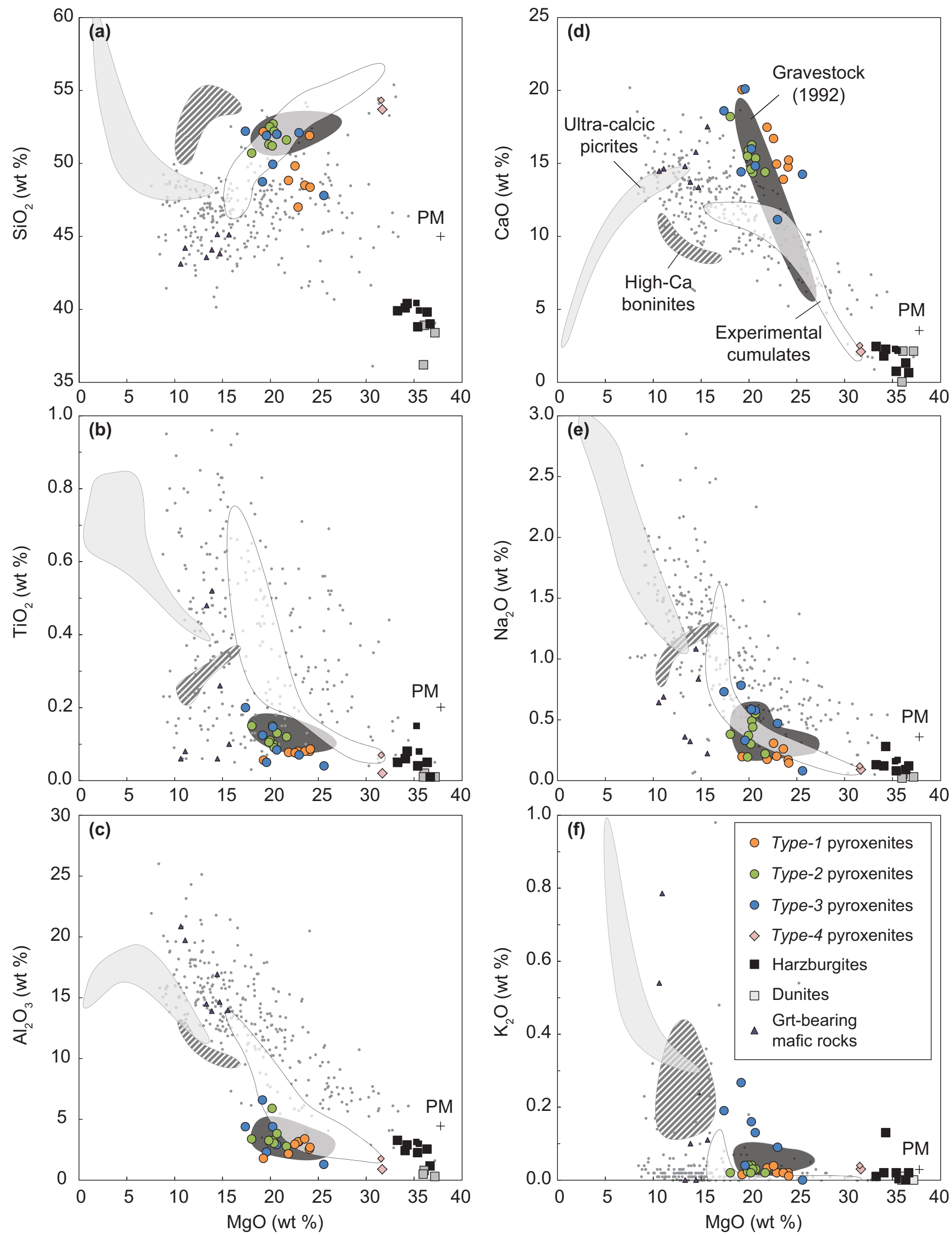

Fig. 7. Variations of whole-rock MgO vs (a) SiO_2, (b) TiO_2, (c) Al_2O_3, (d) CaO, (e) Na_2O and (f) K_2O in the Cabo Ortegal pyroxenites, peridotites and garnet-bearing mafic rocks. Data for peridotites are from Tilhac (2016). Data from Santos *et al.* (2002) are indicated with smaller symbols; the compositional field of Cabo Ortegal pyroxenites from Gravestock (1992) is also shown. Experimental cumulates are pyroxenite compositions reconstructed by Müntener *et al.* (2001) from the crystallization products of hydrous basaltic andesite and high-Mg# andesite at 1·2 GPa. Ultra-calcic picrites are whole-rock compositions of ankaramitic lavas reported by Barsdell & Berry (1990) from Western Epi, Vanuatu. High-Ca boninites are whole-rock compositions of lavas from the Troodos ophiolite, Cyprus and from the north Tonga trench, which were used by Crawford *et al.* (1989) in their classification of boninites. A worldwide database of mantle pyroxenites (massifs and xenoliths) is shown for comparison (Xiong *et al.*, 2014). PM, primitive mantle.

chondritic abundances. Trace element compositions of olivine, opx and garnet can be found in Supplementary Data Electronic Appendix Figs 1 and 2.

The REE patterns of cpx are similar to those of whole-rocks, with ΣREE being slightly lower (Fig. 9). The only exceptions are from garnet-bearing type 2 samples where cpx has negative MREE-to-HREE slopes [$(Gd/Lu)_{CN} > 2$]. Cpx in type 1 pyroxenites commonly has spoon-shaped patterns with $(La/Gd)_N < 1·3$ with few exceptions, whereas it exhibits a range of LREE enrichment in type 2 pyroxenites [$(La/Gd)_{CN} = 0·39–5·8$]. This range is restricted to the highest values for cpx in type 3 pyroxenites [$(La/Gd)_{CN}$ up to 9·1]. In most type 1 pyroxenites and the least LREE-enriched type 2 pyroxenites cpx has flat or slightly negative MREE-to-HREE slopes. In LREE-enriched type 2 and most type 3 pyroxenites it

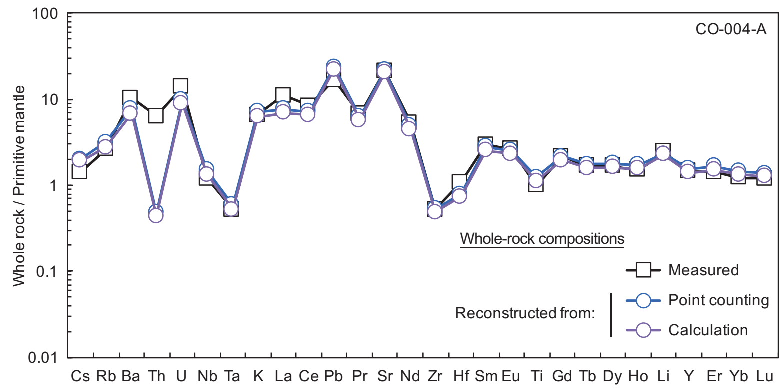

Fig. 8. Comparison between measured whole-rock composition and reconstructions from both point-counted modes and least-squares fitted modes for a type 3 pyroxenite (CO-004-A), plotted on a primitive-mantle-normalized multi-element diagram. Primitive mantle composition from McDonough & Sun (1995).

has fractionated LREE to MREE and may exhibit flat LREE distributions with negative MREE-to-HREE slopes [$(Gd/Lu)_{CN}$ up to 3]. Cpx with the lowest ΣREE contents, particularly in type 2 and 3 pyroxenites, exhibits positive Eu anomalies with $(Eu/Eu^*)_{CN}$ reaching 1·4. LREE-depleted and saddle-shaped REE patterns are observed in cpx from samples CO-006-A and CO-045-A, respectively, which is consistent with their whole-rock compositions. In addition, cpx in one sample of type 4 pyroxenite yielded the highest $(La/Gd)_{CN}$ whereas the other has the most pronounced spoon-shaped pattern among all types. Cpx in a pyroxenite dyke (not included in the main dataset) has a LREE-depleted pattern with the highest HREE content observed (Supplementary Data Electronic Appendix Fig. 1c).

The trace element composition of cpx is characterized by HFSE depletion, particularly in Nb and Ta (Fig. 10), with many analyses below detection limit. Very low Nb results in $(Nb/Ta)_{PM} = 0{\cdot}03$–$2{\cdot}20$, which contrasts with Nb–Ta fractionation in whole-rocks. However, Zr–Hf fractionation is consistent between cpx and whole-rock. Negative Ti anomalies in cpx are well defined in type 3, and to a lesser extent in type 1 pyroxenites, in agreement with the whole-rock data. Cpx from the LREE-enriched opx-rich (type 4) websterite (CO-048) shows even greater anomalies [$(Ti/Ti^*)_{PM} = 0{\cdot}06$; Supplementary Data Electronic Appendix Fig. 2c]. The most incompatible LILE (Cs, Rb, Ba and K) show a wide range of depletion in cpx with respect to PM, whereas Pb and Sr and other fluid-mobile elements such as U and Th are enriched and increase from type 1 to type 2 and 3 pyroxenites. Cpx in the pyroxenite dyke has lower LILE, U, Th and Li than any types of pyroxenite (Supplementary Data Electronic Appendix Fig. 2c).

REE patterns of amphibole are very similar to those of cpx (Fig. 9). However, higher ΣREE results in $D^{\Sigma REE}_{amph\text{-}cpx} \sim 4$–7, with no LREE-to-HREE fractionation. On a multi-element diagram (Fig. 10), a major difference compared with cpx is the enrichment of the most incompatible LILE in amphibole. Rb, Ba and K are particularly strongly partitioned into amphibole; $D_{amph\text{-}cpx}$ are generally well above 15 for those elements. Other LILE and fluid-mobile elements partition either similarly to the REE (U, Th, Pb and Sr) or preferentially into cpx as in the case of Li ($D^{Li}_{amph\text{-}cpx} = 0{\cdot}1$–$0{\cdot}8$). A few exceptions have positive Li anomalies in both amphibole and cpx. HFSE are strongly depleted in amphibole and cpx but, as expected, Nb is preferentially partitioned into amphibole, which results in $D^{Nb}_{amph\text{-}cpx} = 13$–117, whereas $D^{Ta}_{amph\text{-}cpx} = 0{\cdot}7$–$26{\cdot}1$. Ti partitions less preferentially into amphibole than its neighbouring REE, resulting in stronger negative anomalies than in cpx.

Major and trace element profiles

The composition of cpx has been investigated by coupled EMPA–LA-ICP-MS profile analysis of two heterogeneous samples: a composite sample of type 1 and type 3 pyroxenite (CO-010) and a sample that preserves the contact between a type 1 pyroxenite and its host dunite, and thinly layered pyroxenites (CO-013).

Sample CO-010

Along the 8 cm long profile, the modal proportion of amphibole increases steadily from 4 to 14 wt % (Fig. 12a). A selective enrichment of LREE (Fig. 12b), LILE and other fluid-mobile elements (Fig. 13b) accompanies this change in modal composition. This range of LREE compositions [$(La/Gd)_{CH} = 1{\cdot}37$–$19{\cdot}0$] overlaps the entire range observed in type 1 to type 3 pyroxenites. At ~4 cm, the LREE distribution flattens and the enrichment progressively shifts towards the MREE (Fig. 13a). Sr and Pb increase significantly along the profile but not linearly with respect to their neighbouring REE, which results in the transient evolution of $(Pb/Pb^*)_{PM}$ (Fig. 12c) and $(Sr/Sr^*)_{PM}$ (Fig. 12d). The most incompatible LILE, Th and Li remain unchanged or slightly increase. Apart from Li, these elements partition more readily into amphibole and their concentration in cpx may thus respond to small-scale variations in modal proportion. The contribution of olivine

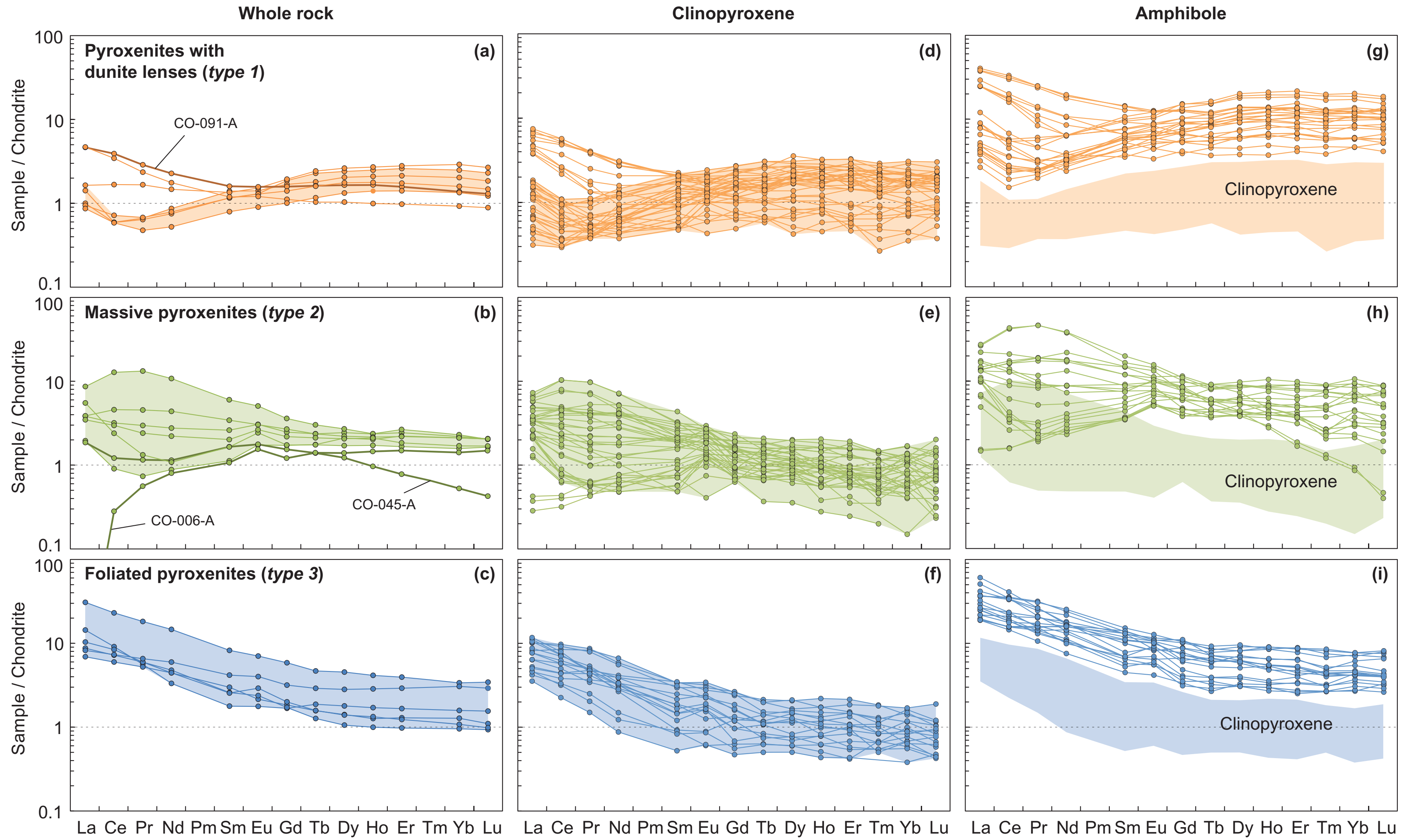

Fig. 9. Chondrite-normalized REE patterns of (a–c) whole-rock samples, (d–f) cpx and (g–i) amphibole from Cabo Ortegal type 1 (a, d, g), type 2 (b, e, h) and type 3 (c, f, i) pyroxenites. Chondrite composition from McDonough & Sun (1995).

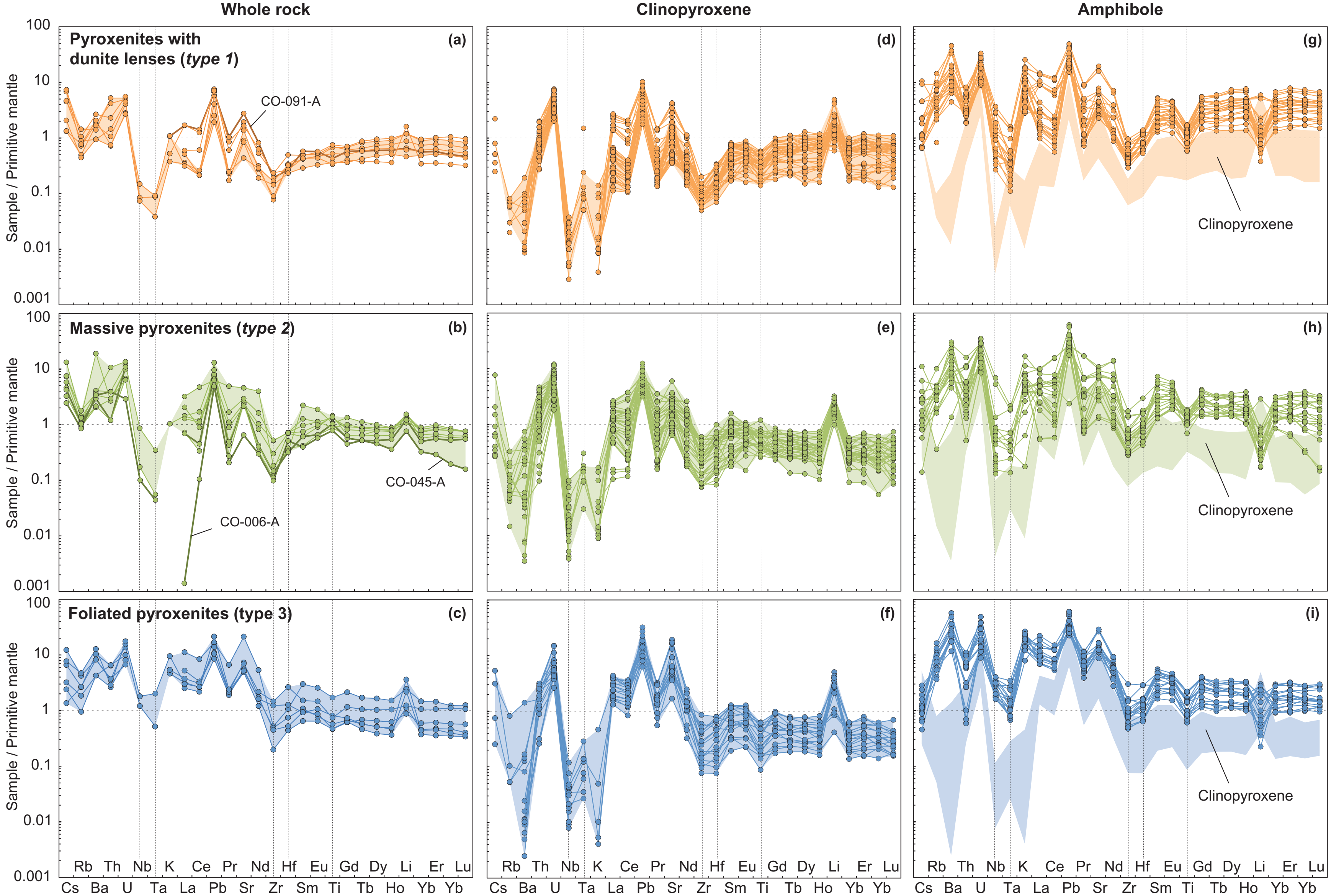

Fig. 10. Primitive-mantle-normalized multi-element patterns of (a–c) whole-rock samples, (d–f) cpx and (g–i) amphibole from Cabo Ortegal type 1 (a, d, g), type 2 (b, e, h) and type 3 (c, f, i) pyroxenites. Primitive mantle composition from McDonough & Sun (1995).

Table 3: Representative major element (wt %) compositions of minerals in Cabo Ortegal pyroxenites

Rock type:	Pyroxenites with dunite lenses (type 1)				Massive pyroxenites (type 2)				Foliated pyroxenites (type 3)				Opx-rich websterites (type 4)		
Mineral:	Cpx														
	amph	I (core)	I (rim)	II	amph	I (core)	I (rim)	II	amph	I (core)	I (rim)	II	I	II	II
SiO_2	53·16	53·20	53·94	53·46	53·18	53·34	53·46	54·31	53·83	54·48	53·12	54·63	54·01	54·32	55·05
TiO_2	0·11	0·03	0·08	0·04	0·19	0·10	0·13	0·10	0·07	0·04	0·04	0·03	0·02	0·01	0·01
Al_2O_3	1·95	1·38	1·07	1·16	2·00	1·75	1·61	1·38	1·10	1·23	0·96	0·91	0·76	0·76	0·47
Cr_2O_3	0·37	0·39	0·35	0·20	0·11	0·33	0·11	0·20	0·42	0·30	0·41	0·28	0·36	0·31	0·23
FeO_t	2·72	2·26	2·32	2·31	3·27	3·33	2·97	3·19	2·27	2·36	3·34	2·36	2·20	2·09	1·97
MgO	17·24	17·67	17·48	17·80	16·61	16·51	16·80	16·55	17·59	17·00	17·57	17·07	17·91	17·83	17·75
NiO	0·04	0·04	0·02	0·05	0·04	0·00	0·00	0·02	0·08	0·02	0·05	0·05	0·04	0·10	0·07
MnO	0·09	0·08	0·06	0·09	0·09	0·15	0·09	0·13	0·09	0·08	0·16	0·10	0·11	0·02	0·08
CaO	24·22	24·89	24·46	24·76	24·29	24·31	24·71	24·04	24·44	24·71	24·20	24·52	24·48	24·40	24·59
Na_2O	0·10	0·10	0·10	0·14	0·09	0·15	0·05	0·17	0·17	0·17	0·37	0·16	0·18	0·18	0·13
K_2O	b.d.l.	b.d.l.	b.d.l.	0·01	b.d.l.	b.d.l.	0·01	0·01	0·01	0·01	0·01	0·01	b.d.l.	b.d.l.	b.d.l.
Total	100·01	100·06	99·90	100·02	99·88	99·96	99·94	100·09	100·07	100·41	100·22	100·12	100·07	100·02	100·34
Mg#	0·92	0·93	0·93	0·93	0·90	0·90	0·91	0·90	0·93	0·93	0·90	0·93	0·93	0·94	0·94
	Opx														
	I (core)	I (core)	I (rim)	II	cpx	I (core)	I (rim)	II	amph	I (core)	I (rim)	II	I (core)	I (rim)	II
SiO_2	55·76	55·73	55·99	56·54	56·07	55·65	56·39	54·22	54·18	53·52	55·30	55·28	55·96	56·86	56·82
TiO_2	0·02	0·04	0·03	0·06	0·05	0·05	0·05	0·03	0·03	0·03	0·03	0·01	0·03	0·01	0·03
Al_2O_3	1·56	2·18	1·48	1·61	1·92	2·38	1·63	2·71	1·83	3·10	1·82	1·51	1·42	1·15	0·76
Cr_2O_3	0·22	0·37	0·17	0·20	0·15	0·30	0·20	0·29	0·34	0·20	0·16	0·18	0·29	0·23	0·23
FeO_t	8·96	7·70	9·11	8·49	10·61	11·21	12·04	12·96	11·45	11·02	10·98	10·05	7·57	7·67	7·76
NiO	0·03	0·02	0·01	0·08	0·05	0·00	0·01	0·01	0·03	0·00	0·03	0·04	0·08	0·05	0·09
MnO	0·25	0·21	0·29	0·22	0·15	0·33	0·37	0·27	0·32	0·28	0·26	0·22	0·19	0·22	0·16
MgO	33·30	33·41	33·14	33·26	31·53	29·96	29·50	29·49	31·68	31·80	31·83	32·55	34·18	34·36	33·77
CaO	0·26	0·56	0·25	0·21	0·24	0·25	0·26	0·21	0·24	0·23	0·26	0·24	0·24	0·22	0·33
Na_2O	b.d.l.	b.d.l.	b.d.l.	b.d.l.	0·03	0·01	0·01	b.d.l.	b.d.l.	b.d.l.	b.d.l.	0·01	b.d.l.	0·01	0·01
K_2O	b.d.l.	b.d.l.	0·01	0·01	0·01	b.d.l.	b.d.l.	b.d.l.	0·02	0·02	0·01	0·01	b.d.l.	b.d.l.	0·02
Total	100·37	100·21	100·47	100·68	100·81	100·13	100·45	100·18	100·12	100·18	100·65	100·12	99·96	100·79	99·99
Mg#	0·87	0·89	0·87	0·88	0·84	0·83	0·81	0·80	0·83	0·84	0·84	0·85	0·89	0·89	0·89
	Amph														
	idio	sub	xeno	encl	idio	sub	xeno	encl	idio	sub	xeno	encl	xeno	sub	sub
SiO_2	46·68	49·16	46·70	47·41	48·38	47·04	49·53	47·65	45·63	48·40	47·70	49·46	48·06	54·64	49·16
TiO_2	0·26	0·22	0·26	0·22	0·32	0·28	0·36	0·36	0·27	0·32	0·16	0·33	0·15	0·04	0·12
Al_2O_3	10·18	9·03	9·90	8·17	10·13	10·78	8·62	10·21	10·10	8·95	9·85	8·43	9·28	3·29	8·31
Cr_2O_3	0·77	0·55	1·11	0·71	1·09	0·65	0·76	0·30	0·72	0·88	1·00	0·57	1·25	0·33	0·96
FeO_t	4·74	4·39	4·22	3·71	5·76	5·67	5·36	5·32	6·26	7·60	4·30	7·01	4·28	3·05	4·10
NiO	0·06	0·05	0·10	0·10	0·06	0·02	0·05	0·04	0·03	0·00	0·05	0·01	0·11	0·05	0·03
MnO	0·08	0·06	0·08	0·06	0·08	0·01	0·00	0·05	0·10	0·09	0·05	0·09	0·06	0·06	0·10
MgO	19·69	19·76	19·65	21·31	18·20	18·63	19·08	17·50	19·06	17·24	18·84	18·14	19·78	22·55	19·97
CaO	12·69	12·71	12·82	12·82	12·27	12·36	12·73	12·73	12·61	12·39	12·79	12·35	12·44	12·82	12·60
Na_2O	1·81	1·16	1·77	1·63	1·53	1·85	1·12	1·27	1·62	1·44	1·67	1·34	1·94	0·56	1·52
K_2O	0·26	0·22	0·33	0·17	0·08	0·20	0·19	0·08	0·43	0·51	0·47	0·49	0·13	0·13	0·54
Total	97·21	97·31	96·94	96·30	97·90	97·50	97·78	95·52	96·84	97·82	96·90	98·23	97·47	97·53	97·43
Mg#	0·88	0·89	0·89	0·91	0·85	0·85	0·86	0·85	0·84	0·80	0·89	0·82	0·89	0·93	0·90

	Ol				Grt			
					core	rim	coro	
SiO_2	40·06	40·11	39·89	40·04	39·84	40·48	40·38	40·16
TiO_2	b.d.l.	0·02	0·02	b.d.l.	0·10	0·14	0·02	0·02
Al_2O_3	b.d.l.	b.d.l.	0·02	0·01	21·48	21·24	22·57	22·27
Cr_2O_3	b.d.l.	0·03	0·01	0·02	1·09	1·01	0·19	0·66
FeO_t	11·57	13·32	10·99	11·76	16·15	16·35	15·27	15·61
NiO	0·15	0·23	0·29	0·23	0·00	0·00	0·06	0·01
MnO	0·22	0·24	0·21	0·21	1·03	0·95	0·82	0·94
MgO	48·02	46·10	48·22	48·15	12·57	12·44	14·68	14·48
CaO	0·03	0·03	0·03	0·03	7·90	7·82	6·03	6·14
Na_2O	0·02	0·01	0·02	b.d.l.	0·01	0·00	0·01	0·01
K_2O	b.d.l.	b.d.l.	0·04	0·01	0·01	0·00	0·02	b.d.l.
Total	100·08	100·14	99·71	100·44	100·18	100·44	100·04	100·31
Mg#	0·88	0·86	0·89	0·88	0·58	0·58	0·63	0·62

Amph, abundant amphibole along cleavage (or after exsolved cpx); idio, idiomorphic; sub, sub-idiomorphic; xeno, xenomorphic; encl., enclosed; coro, corona around spinel (see text for details on the petrographic features); FeO_t, total Fe as FeO; b.d.l., below detection limit; Mg#, molar ratio of $Mg/(Mg + Fe^{tot})$.

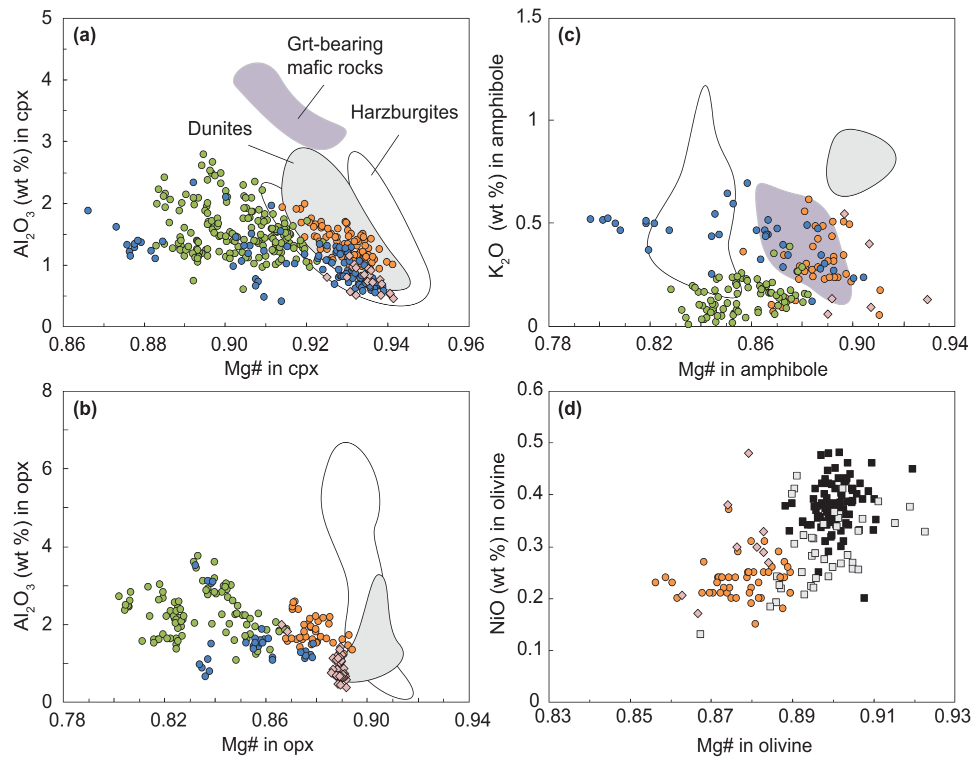

Fig. 11. Mineral compositions in Cabo Ortegal pyroxenites, peridotites and the garnet-rich mafic layer near the top of the pyroxenite-rich body: (a) Mg# vs Al_2O_3 in cpx; (b) Mg# vs Al_2O_3 in opx; (c) Mg# vs K_2O in amphibole; (d) Mg# vs NiO in olivine. Data for harzburgites, dunites and the garnet-rich mafic layer near the top of the pyroxenite-rich body are from Santos *et al.* (2002). Symbols as in Fig. 7.

(±opx) is also suggested by the variations in compatible elements such as Ni and Co (Fig. 12e), whereas Cr content is buffered by the presence of Cr-spinel throughout the profile.

Sample CO-013

The 12 cm long profile can be described in terms of two distinct trends. From 0 to 7 cm, which corresponds to the type 1 pyroxenite itself, the modal proportion of amphibole decreases from 12 to 7 wt % (Fig. 14a) along with the lightest REE (La to Nd; Fig. 14c and d). From 7 to 12 cm, which includes the contact between the main pyroxenite and thinly layered pyroxenites and dunite, the modal proportion of amphibole decreases from 7 to 5 wt %, whereas serpentinization increases dramatically from ~30 to ~70%. Within this section, Li increases significantly (Fig. 14b), from 1·6 to 7·0 ppm [$(Li/Dy)_{PM}$ = 1·1–5·3] along with the lightest REE only (La and Ce). Ni (Fig. 14e) and, to a lesser extent, Co exhibit similar small-scale variations related to the evolution of the modal proportion of olivine to those described for sample CO-010.

DISCUSSION

Impact of secondary processes

In the Herbeira massif of the Cabo Ortegal Complex, pyroxenite layers vary in thickness from a few millimetres to 3 m. This may reflect inheritance of intrusive magmatic layering strongly overprinted by tectonic slicing and transposition, as suggested by Girardeau & Gil Ibarguchi (1991). Cabo Ortegal pyroxenites also show evidence of prograde metamorphism from the stability field of spinel pyroxenite to that of garnet pyroxenite and of hydration during retrograde metamorphism, notably under amphibolite-facies conditions, and serpentinization. Selective enrichment of LILE, LREE and other fluid-mobile elements over the HFSE is also characteristic of these pyroxenites and may relate to fluid or melt percolation. A preliminary requirement is thus to assess the impact of these secondary processes in order to envision the magmatic significance of Cabo Ortegal pyroxenites, based on the consistency between the petrological and geochemical systematics.

Serpentinization

Cabo Ortegal pyroxenites, particularly olivine-bearing (type 1) pyroxenites, have been subjected to various degrees of serpentinization. To investigate the role of this process, the composition of cpx was analysed across the contact between a type 1 pyroxenite (CO-013; Fig. 14) and its host dunite (and thinly layered pyroxenites). With serpentinization increasing from ~30 to ~70%, Li increases significantly (Fig. 14b) along with the lightest REE (La and Ce). This very probably represents the diffusional modification of a sharp gradient, formerly

Table 4: Representative trace element (ppm) compositions of cpx in Cabo Ortegal pyroxenites

Rock type:	Type 1 pyroxenites			Type 2 pyroxenites			Type 3 pyroxenites			Type 4 pyroxenites	
	I (core)	I (rim)	II	amph	I (core)	I (rim)	amph	I (core)	I (rim)	I (core)	I (core)
Li	2·7	1·7	0·9	3·4	2·8	3·6	1·2	6·1	0·9	3·6	2·3
Be	b.d.l.	b.d.l.	0·16	0·14	0·43	0·40	0·15	1·04	0·10	0·60	b.d.l.
Sc	50	53	65	73	47	59	64	85	56	59	50
V	116	104	149	214	136	118	113	196	84	85	106
Cr	2340	1970	1550	2130	820	1300	1140	1780	850	3570	1560
Co	21	18	17	26	20	23	21	22	18	23	21
Ni	162	187	166	103	78	110	185	84	141	n.a.	247
Cu	b.d.l.	b.d.l.	0·1	b.d.l.	b.d.l.	0·1	5·8	0·2	0·1	b.d.l.	b.d.l.
Zn	5·6	2·8	2·8	14·6	9·2	12·6	5·2	20·3	3·8	n.a.	5·4
Ga	1·23	0·84	0·92	2·53	2·26	1·48	1·27	3·06	0·75	1·09	0·68
Rb	b.d.l.	b.d.l.	b.d.l.	0·10	b.d.l.	0·03	0·25	0·06	0·01	b.d.l.	b.d.l.
Sr	39	20	80	88	42	78	111	376	103	166	28
Y	0·90	2·94	2·43	1·50	1·10	1·31	1·87	2·36	1·18	1·68	1·25
Zr	0·8	1·2	1·3	3·5	3·5	1·1	1·6	4·0	1·0	1·8	0·3
Nb	0·01	b.d.l.	0·01	0·03	0·01	0·02	0·15	0·02	b.d.l.	b.d.l.	b.d.l.
Cs	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	b.d.l.	0·11	b.d.l.	b.d.l.	b.d.l.
Ba	b.d.l.	0·1	0·1	n.a.	0·2	0·4	13·0	0·5	b.d.l.	0·1	b.d.l.
La	0·41	0·92	1·45	0·95	0·83	1·04	2·77	1·92	1·94	4·78	0·16
Ce	0·59	1·34	3·08	2·96	2·18	3·95	4·59	5·15	3·26	11·54	0·14
Pr	0·06	0·14	0·37	0·47	0·30	0·60	0·44	0·76	0·31	1·43	0·02
Nd	0·23	0·59	1·25	2·51	1·48	2·30	1·35	3·02	0·87	4·87	0·08
Sm	0·07	0·25	0·33	0·55	0·27	0·37	0·17	0·50	0·19	0·69	0·04
Eu	0·05	0·06	0·10	0·19	0·08	0·11	0·09	0·19	0·05	0·17	0·03
Gd	0·12	0·28	0·33	0·40	0·18	0·32	0·21	0·52	0·14	0·34	0·12
Tb	0·02	0·06	0·06	0·06	0·03	0·04	0·05	0·07	0·03	0·06	0·02
Dy	0·19	0·49	0·48	0·38	0·15	0·19	0·28	0·40	0·24	0·24	0·24
Ho	0·04	0·10	0·10	0·06	0·03	0·06	0·07	0·09	0·04	0·06	0·05
Er	0·10	0·30	0·31	0·12	0·09	0·14	0·19	0·24	0·15	0·16	0·13
Tm	0·02	0·05	0·03	0·01	0·01	0·02	0·04	0·03	0·02	0·02	0·02
Yb	0·11	0·28	0·26	b.d.l.	0·14	0·18	0·18	0·26	0·11	0·18	0·15
Lu	0·02	0·05	0·04	0·01	0·01	0·02	0·03	0·03	0·02	0·02	0·02
Hf	0·05	0·08	0·04	0·22	0·12	0·03	0·06	0·18	0·05	0·04	b.d.l.
Ta	0·003	b.d.l.	b.d.l.	b.d.l.	0·005	b.d.l.	0·004	0·001	b.d.l.	b.d.l.	b.d.l.
Pb	1·1	0·6	1·4	1·3	1·0	0·9	1·9	2·4	1·5	2·8	1·2
Th	0·06	0·11	0·09	0·03	0·22	0·05	0·24	0·02	0·10	0·14	0·01
U	0·11	0·09	0·09	0·09	0·08	0·10	0·13	0·12	0·07	0·06	0·01

Amph, abundant amphibole along cleavage (see text for details on the petrographic features); b.d.l., below detection limit; n.a., not analyzed.

located at the contact between moderately serpentinized pyroxenite and highly serpentinized dunite. Additional enrichment of highly fluid-mobile elements can be thus potentially ascribed to serpentinization but it seems to be restricted to the lightest REE and Li, as observed only in certain type 1 pyroxenites.

Fluid or melt percolation

Cpx, amphibole and whole-rock samples consistently show REE patterns ranging from spoon-shaped in (type 1) pyroxenites with dunite lenses, to variously LREE-enriched and LREE-to-MREE fractionated in massive (type 2) and foliated (type 3) pyroxenites (Fig. 9). Gravestock (1992) noted that, among melting models, only fractional disequilibrium melting could produce such selective enrichment of LREE (Prinzhofer & Allègre, 1985; Bédard, 1989). However, this would result in a positive MREE-to-HREE slope, which is not observed in Cabo Ortegal pyroxenites. The range of spoon-shaped to continuously enriched REE patterns can instead be reproduced by percolation of fluids or melts according to the chromatographic model of Navon & Stolper (1987), as reported in many localities (e.g. Bodinier *et al.*, 1990; O'Reilly *et al.*, 1991; Takazawa *et al.*, 1992; Batanova *et al.*, 1998; Godard & Martin, 2000; Ionov *et al.*, 2002; Powell *et al.*, 2004; Gysi *et al.*, 2011). This can account for the progressive enrichment of LREE and MREE observed within sample CO-010, and also between the different types of pyroxenite, as chromatographic re-equilibration can potentially generate extreme fractionation at small scales (e.g. Rampone & Morten, 2001). Accordingly, the deviation of Ce or Nd from a simple mixing line when plotted against Sm (Fig. 15) and the transient evolution of $(Pb/Pb^*)_{PM}$ and $(Sr/Sr^*)_{PM}$ (Fig. 12c and d) are predicted between elements of different degrees of incompatibility by the chromatographic model (Navon & Stolper, 1987).

Potential metasomatic agents responsible for the chromatographic re-equilibration of Cabo Ortegal pyroxenites include residual melts left after fractional crystallization and hydrous metamorphic fluids. These alternatives have very distinctive petrogenetic implications closely related to the origin of amphibole and are

Table 5: Representative trace element (ppm) compositions of amphibole in Cabo Ortegal pyroxenites

Rock type:	Type 1 pyroxenites			Type 2 pyroxenites			Type 3 pyroxenites			Type 4 pyroxenites	
	idio	sub	sub	xeno	idio	xeno	idio	sub	sub	xeno	sub
Li	1·0	0·6	1·3	1·2	0·6	4·6	1·6	3·5	0·6	9·2	15·5
Be	0·07	0·12	0·09	b.d.l.	0·11	0·46	1·45	0·82	0·35	0·16	1·25
Sc	110	169	115	118	101	55	99	73	107	112	98
V	402	577	375	386	480	289	428	371	487	324	182
Cr	5862	2679	5555	5032	5573	1638	3794	5094	4113	7471	6097
Co	41	38	33	65	42	18	47	45	47	39	34
Ni	451	508	519	359	569	115	248	459	272	734	553
Cu	0·1	0·125	3·5	0·117	0·1	0·1	0·2	0·5	0·1	0·084	0·192
Zn	9·4	7·1	5·0	20·9	9·0	9·3	46·1	32·0	22·4	9·4	10·2
Ga	5·23	3·84	4·53	6·60	7·69	1·94	11·14	6·86	9·40	4·64	4·76
Rb	3·2	2·1	4·4	1·6	2·0	0·2	5·5	3·6	4·5	3·0	2·5
Sr	122	47	69	27	62	50	577	250	309	57	416
Y	5·61	25·17	13·83	6·89	4·79	6·57	12·26	4·11	7·42	6·44	7·08
Zr	3·5	5·3	3·6	3·3	2·9	6·9	8·9	5·6	10·1	2·1	13·5
Nb	0·77	0·27	0·62	0·12	0·14	0·41	2·57	1·24	2·04	0·10	2·81
Cs	0·02	0·05	0·04	0·01	0·02	0·06	0·01	0·02	0·01	0·04	b.d.l.
Ba	73	81	133	78	44	9	170	114	237	50	126
La	2·11	1·88	1·03	0·35	2·33	3·13	8·69	6·28	4·48	0·74	22·18
Ce	2·76	3·52	1·59	0·96	2·39	7·88	21·21	11·88	9·71	0·65	51·52
Pr	0·29	0·56	0·21	0·20	0·22	1·08	2·85	1·32	1·47	0·07	6·23
Nd	1·09	2·92	1·52	1·38	1·05	6·05	11·57	5·15	7·15	0·48	21·82
Sm	0·56	1·39	0·89	0·52	0·54	1·30	2·25	0·85	1·93	0·28	3·14
Eu	0·19	0·58	0·35	0·28	0·29	0·49	0·72	0·31	0·59	0·11	0·83
Gd	0·83	2·70	1·38	0·90	0·76	0·95	2·06	0·78	1·73	0·50	1·85
Tb	0·13	0·55	0·30	0·17	0·14	0·14	0·33	0·12	0·26	0·11	0·20
Dy	0·93	4·38	2·45	1·09	0·97	0·93	2·35	0·75	1·54	0·99	1·24
Ho	0·20	0·99	0·54	0·28	0·22	0·20	0·47	0·16	0·30	0·25	0·26
Er	0·66	2·94	1·45	0·77	0·63	0·50	1·42	0·44	0·82	0·86	0·75
Tm	0·09	0·42	0·23	0·12	0·06	0·06	0·19	0·07	0·11	0·13	0·11
Yb	0·75	2·76	1·57	0·93	0·60	0·49	1·13	0·43	0·73	0·94	0·82
Lu	0·10	0·38	0·24	0·12	0·08	0·06	0·20	0·08	0·10	0·17	0·15
Hf	0·18	0·38	0·23	0·14	0·12	0·29	0·31	0·19	0·46	0·12	0·47
Ta	0·010	b.d.l.	0·016	b.d.l.	b.d.l.	0·012	0·061	0·038	0·088	b.d.l.	0·104
Pb	4·5	3·3	3·1	2·6	3·7	0·4	4·9	3·8	8·2	2·6	5·0
Th	0·20	0·23	0·20	0·04	0·83	0·46	0·06	0·25	0·49	0·05	0·73
U	0·46	0·39	0·24	0·17	0·59	0·21	0·34	0·25	0·77	0·10	0·43

Idio, idiomorphic; sub, sub-idiomorphic; xeno, xenomorphic; b.d.l., below detection limit.

discussed below. Alternatively, the migration of low-degree partial melts could be envisaged assuming that small-volume melts are able to percolate through a solid matrix (McKenzie, 1984). Post-kinematic migration of low-degree partial melts may indeed be indicated by pyroxenite and garnet-rich mafic dykes in the Herbeira massif. Although the presence of such dykes in the vicinity of LILE- and LREE-enriched type 1 pyroxenites is demonstrated for one sample (CO-092-A), we believe that this non-pervasive process cannot be responsible for the ubiquitous metasomatic signature of the Cabo Ortegal pyroxenites. In addition, the very limited enrichment of LREE, LILE and other fluid-mobile elements observed in these dykes (CO-064-C; Supplementary Data Electronic Appendix Figs 1c and 2c) confirms that their intrusion post-dated the main magmatic–metasomatic episode.

Metamorphic vs late magmatic–metasomatic amphibolitization

The origin of amphibole, which represents at the thin-section scale between 4 and 42 wt % of the Cabo Ortegal pyroxenites, is ambiguous. There are two scenarios, not mutually exclusive, according to which amphibole represents the product of (1) hydration (± addition of volatiles) of cpx and spinel during retrograde metamorphism and/or (2) crystallization of percolating or trapped residual melts.

The first scenario is supported by textural observations, and particularly the post-kinematic character of amphibole and its dominantly interstitial position, which clearly indicate that it is, at least partly, a secondary phase. It is also consistent with the observation that cpx and opx from the most amphibolitized (type 3) pyroxenites overlap the range of Mg# of both type 1 and type 2 and have lower Al_2O_3 contents owing to subsolidus re-equilibration with amphibole (Fig. 11a and b). The partitioning of most incompatible elements (apart from REE, as discussed below) is similar to values reported from xenoliths (Vannucci *et al.*, 1995; Chazot *et al.*, 1996; Ionov *et al.*, 1997; Grégoire *et al.*, 2000*b*). This may indicate chemical equilibrium between the two phases, in good agreement with the similarities between their REE patterns (Fig. 9), which is in turn consistent with a metamorphic origin for amphibole. Type

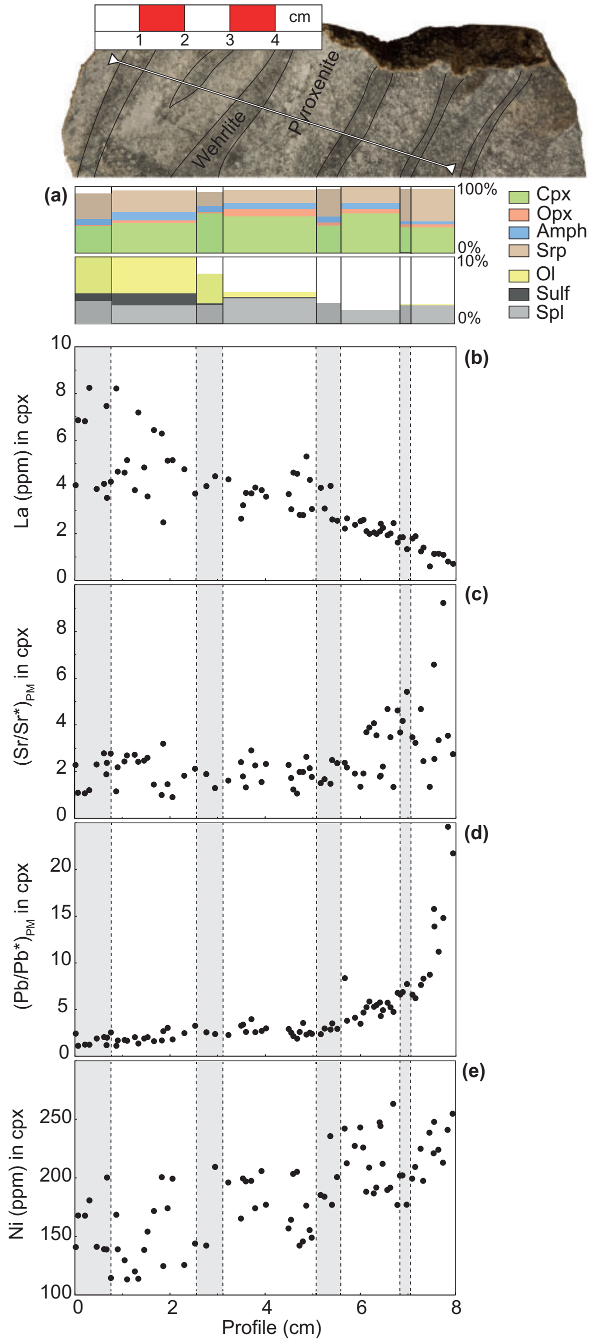

Fig. 12. Evolution of (a) modal composition and (b–e) composition of cores of cpx I and of a few neoblasts (cpx II) across a composite sample of type 1 and type 3 pyroxenite (CO-010): (b) La; (c) $(Sr/Sr^*)_{PM}$; (d) $(Pb/Pb^*)_{PM}$; (e) Ni. (Note the wehrlite streak preserved within the thick pyroxenite layer corresponding to former dunite lenses; see text for further details.)

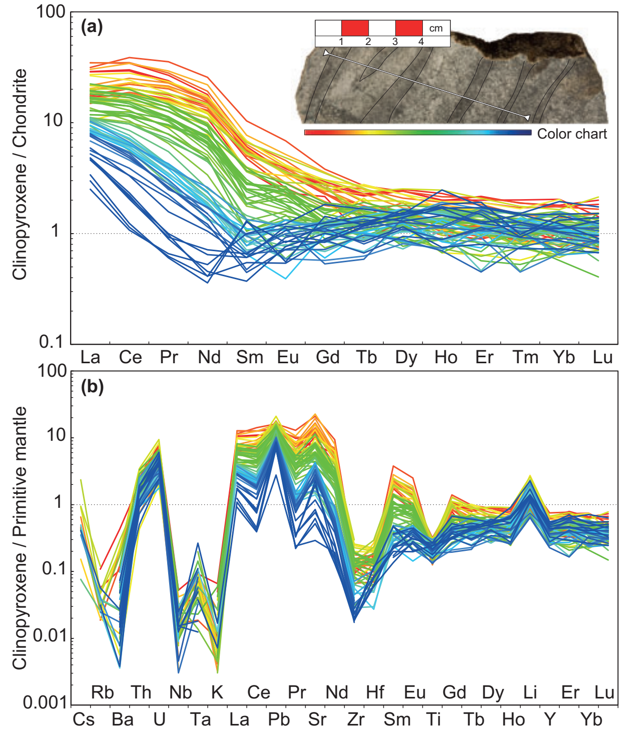

Fig. 13. (a) Chondrite-normalized REE patterns and (b) primitive-mantle-normalized multi-element patterns of cpx along the profile of Fig. 12 (also reproduced here). Colour code relates to positions along the profile. Chondrite and primitive mantle compositions from McDonough & Sun (1995).

3 pyroxenites may indeed simply have formed through greater amphibolitization (i.e. hydration) of the same protoliths as those of type 1 and 2 pyroxenites. This implies that more deformed protoliths were subsequently exposed to greater hydration, and hence amphibolitization. In addition, type 3 pyroxenites plot parallel to the Fo–An join following type 1 pyroxenites on a pseudoternary forsterite–Ca-Tschermak pyroxene–quartz (Fo–CaTs–Qz) projection (Fig. 16), which suggests that primitive olivine-clinopyroxenitic and olivine-websteritic protoliths were preferentially amphibolitized, in agreement with textural relationships in type 1 pyroxenites (Fig. 4e and f). Conversely, the preservation of dunite lenses within them is indicative of limited metasomatic–metamorphic overprint and deformation.

Alternatively, several observations support the second scenario. Strong partitioning of the REE into amphibole ($D^{\Sigma REE}_{amph\text{-}cpx} \sim 4$–7) as measured in Cabo Ortegal pyroxenites has been reported in only a few other localities (Witt-Eickschen & Harte, 1994; Vannucci *et al.*, 1995). In the East Eifel xenoliths of the Rhenish Massif, similar high values of $D^{\Sigma REE}_{amph\text{-}cpx}$ (3–8) were interpreted as the result of a crystal-chemical control from low Na and K, and high Ca in cpx, similar to cpx compositions in Cabo Ortegal pyroxenites. In these xenoliths, amphibolitization has been attributed to the interaction of rising peridotites with small fractions of low-density, CO_2–H_2O-bearing fluids present along grain boundaries during cooling, without additional supply of percolating fluids (Witt & Seck, 1989). We suggest that the presence of small volumes of incompatible- and volatile-enriched residual melts after segregation of the Cabo Ortegal pyroxenites is responsible for the late magmatic–metasomatic crystallization of hornblende, as argued by Harte *et al.* (1993). This has been previously proposed for similar bulk compositions

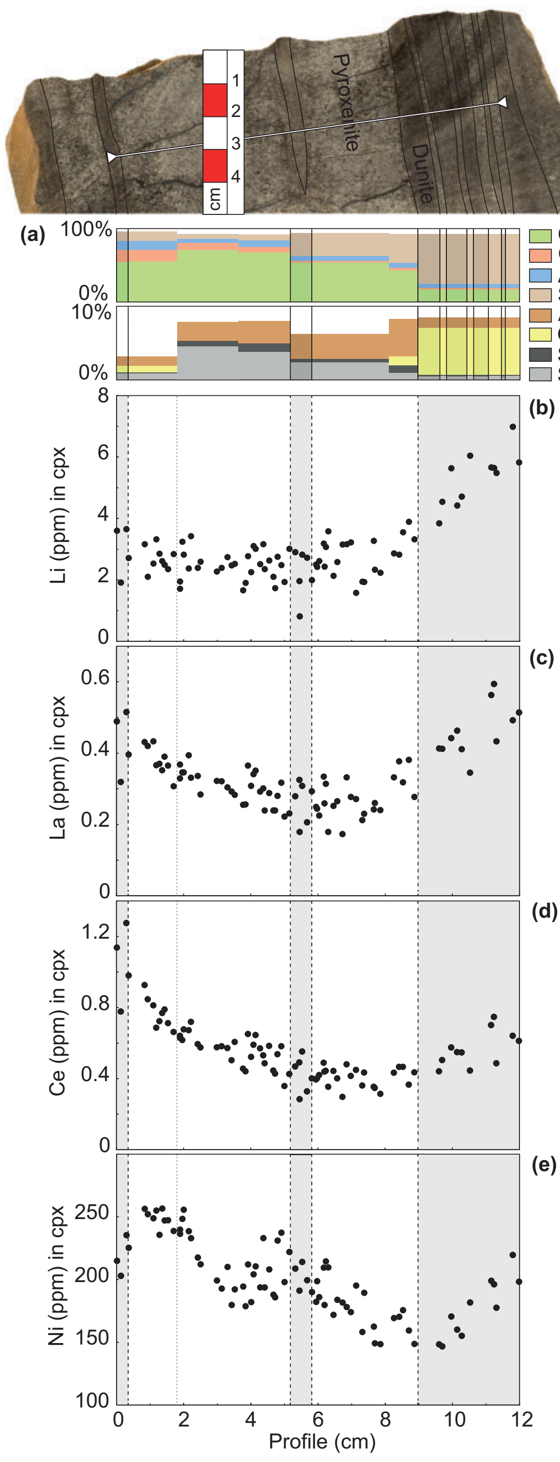

Fig. 14. Evolution of modal composition (a) and composition of cores of cpx I and of a few neoblasts (cpx II) across a sample of type 1 pyroxenite and its host dunite and thinly layered pyroxenites (CO-013): (b) Li; (c) La; (d) Ce; (e) Ni. (Note the dunite lens preserved within the pyroxenite layer as in Fig. 12; see text for further details.)

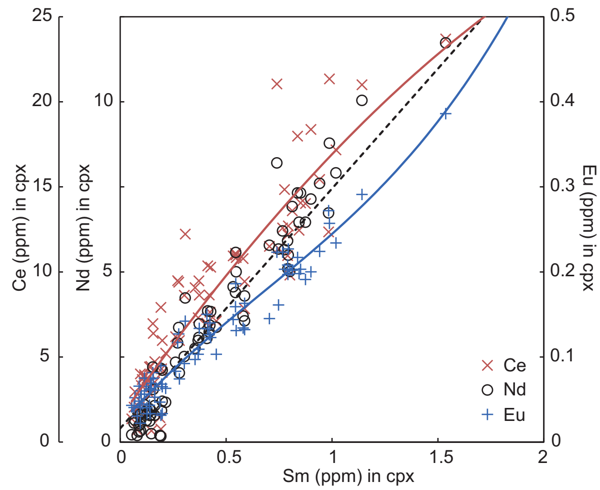

Fig. 15. Concentration of Ce, Nd and Eu in cpx from the profile shown in Fig. 12 plotted vs Sm. Scales on the *y*-axis are adjusted so that end-member samples plot at approximately the same position, as in the study by Navon & Stolper (1987). Red and blue lines represent best-fit regression lines for Ce and Eu, respectively. Dashed black line represents the linear trend expected from simple mixing between end-member samples.

(Green *et al.*, 2004) and documented in other arc-related pyroxenites (Burg *et al.*, 1998). Such metasomatic infill would account for the correlation between the modal proportion of amphibole and the selective enrichment of incompatible trace elements described above, owing to the combination of porosity-reducing reactions (fractional crystallization) and chromatographic effects (Navon & Stolper, 1987; Bodinier *et al.*, 1990; Vasseur *et al.*, 1991), referred to as 'percolative fractional crystallization' by Harte *et al.* (1993). It is also expected from experimental studies that a compositional shift from H_2O-rich to CO_2-rich in the residual liquid would result from amphibole precipitation and consequently reduce the solubility of the REE, which would thus be incorporated into cpx and amphibole (Schneider & Eggler, 1986).

For these reasons, we believe that the existence of late magmatic–metasomatic amphibole is very likely. However, compelling textural evidence supports the existence of at least some metamorphic amphibole and thus suggests two distinct episodes of amphibolitization (i.e. late magmatic–metasomatic and metamorphic). Nonetheless, both episodes may be regarded as occurring in virtually closed systems, which allows us to speculate on the petrogenetic processes and on the nature of melts involved.

Origin of Cabo Ortegal pyroxenites

Girardeau & Gil Ibarguchi (1991) estimated that 80–90% of the 300 m thick, 3 km long pyroxenite-rich body exposed in the Herbeira massif of the Cabo Ortegal Complex is made up of pyroxenites, which is consistent with our observations. This amount of relatively homogeneous pyroxenite requires a significant volume of compositionally similar melts. It is unlikely that these were produced *in situ* (i.e. in a closed-system) by low-degree partial melting of lherzolite, as previously proposed (Van Calsteren, 1978; Van Calsteren *et al.*, 1979). Instead, high-pressure crystallization of an external

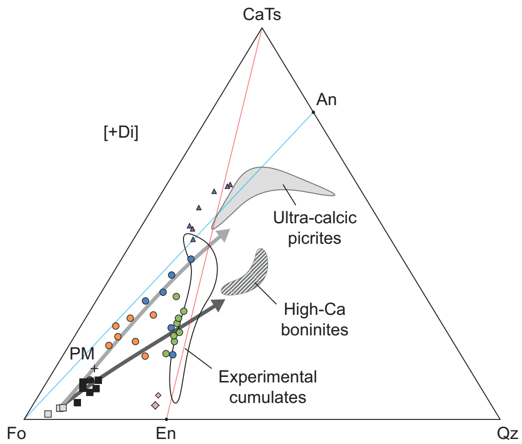

Fig. 16. Normative compositions of Cabo Ortegal pyroxenites in the pseudoternary system Fo–CaTs–Qz projected from Di using the method of O'Hara (1968). Nepheline- and hypersthene-normative compositions are separated by the Di–Fo–An plane (blue line). Silica-deficient and silica-excess compositions are divided by the Di–CaTs–En plane (red line). Light grey and dark grey arrows correspond to the potential differentiation trends for type 1 pyroxenites and type 2 pyroxenites, towards picrites and boninites, respectively (see text for further details). Fo, forsterite; CaTs, Ca-Tschermak pyroxene; Qz, quartz; An, anorthite; En, enstatite; Di, diopside. Symbols as in Fig. 7.

melt intruding and reacting with the ambient peridotite has been suggested more recently (Girardeau *et al.*, 1989; Girardeau & Gil Ibarguchi, 1991; Gravestock, 1992) and appears to be a more reasonable scenario. More recently, Moreno *et al.* (2001) interpreted the western part of the Herbeira massif as a (crustal) cumulate sequence to account for high-Pd and -Pt chromitites exposed within high-Cr# dunites. Santos *et al.* (2002) confirmed the residual origin of the harzburgites and showed a fractionation trend between dunites and pyroxenites. However, a comprehensive model accounting for the formation of the Cabo Ortegal pyroxenites based on petrological and geochemical constraints is still missing. This is thus the aim of the following discussion.

We suggest that the lithological variations observed among the Cabo Ortegal pyroxenites (e.g. olivine clinopyroxenites, clinopyroxenites, websterites and opx-rich websterites), and their association with dunites, relate to an intricate combination of magmatic differentiation and melt–peridotite interaction. For that purpose, several assumptions have been made: (1) all pyroxenites are cogenetic and related to a single partial melting episode of a homogeneous mantle source, in agreement with homogeneous Pb-, Sr- and Nd-isotope compositions previously reported (Gravestock, 1992; Santos *et al.*, 2002); (2) they do not represent the metamorphic products of plagioclase-bearing mafic lithologies, as suggested by the absence of strong positive Eu anomaly; (3) their whole-rock major element compositions were preserved ($\pm$ volatiles addition) during amphibolitization, as discussed above.

Major element characteristics of the parental melts

The whole-rock major element compositions of the Cabo Ortegal pyroxenites are characterized by very high CaO and low Al_2O_3, TiO_2 and alkalis. They are thus comparable with the pyroxenitic compositions reconstructed by Müntener *et al.* (2001) from the experimental products of crystallization of hydrous basaltic andesite and high-Mg# andesite at 1·2 GPa. The corresponding experimental cpx have low Ca- and Cr-Tschermaks (CaTs + CrTs) indicative of near-liquidus crystallization where the partitioning of Al between cpx and melt increases with water content and decreases with temperature (Gaetani *et al.*, 1993; Sisson & Grove, 1993; Müntener *et al.*, 2001). CaTs + CrTs components may then increase during fractional crystallization owing to the combined effect of high pressure and high water content in the melt (>3%) preventing the appearance of plagioclase. This trend is observed in the cpx of Cabo Ortegal pyroxenites and arc-related pyroxenites and gabbronorites from the Tonsina complex (DeBari & Coleman, 1989), but their wollastonite (Wo) content is shifted away from the compositional field of experimental cpx (Supplementary Data Electronic Appendix Fig. 3). This shift is ascribed to the widening of the immiscibility gap between enstatite and diopside at lower temperature, owing to sub-solidus re-equilibration and/or the presence of water during crystallization (Gaetani & Grove, 1998; Müntener *et al.*, 2001). Accordingly, our data show that cpx in type 1 and 4 pyroxenites has slightly higher Wo contents than that in type 2 pyroxenites, potentially owing to sub-solidus re-equilibration with higher modal olivine.

In addition to the effect of pressure and water content, experiments have shown that, provided that there are no Al-rich phases present, a positive correlation exists between the partitioning of Al into cpx and the Al/Si of the coexisting melt (Gaetani & Grove, 1995; Müntener *et al.*, 2001). Low-Al pyroxenes such as reported in this study are thus likely to be produced from hydrous melts with low Al/Si such as boninites or high-Mg# andesites, as suggested by Müntener *et al.* (2001) for two-pyroxene saturated melts. However, we believe that Si-undersaturated melts such as picrites cannot be excluded at this stage owing to the uncertain extent of olivine fractionation and melt–rock interaction. We emphasize that the terms boninite and picrite are here strictly used to describe the affinities of their parental melts, and not the Cabo Ortegal pyroxenites themselves. They were defined on a petrogenetic basis by Crawford *et al.* (1989) for describing magmatic rocks 'demonstrably derived from parental magmas meeting [...] compositional requirements', which for boninites are high SiO_2 (>52 wt %) and MgO (>8 wt %) and low TiO_2 (<0·5 wt %), and for picrites are high MgO (>12 wt %), and low SiO_2 (<52%) and alkalis (<3 wt % $Na_2O + K_2O$), as revised by Le Bas (2000). The very high CaO/Al_2O_3 (2·2–11·3) of the Cabo Ortegal pyroxenites more specifically suggests high-Ca boninites (Crawford *et al.*, 1989) or ultra-calcic picrites (Schiano *et al.*, 2000; Kogiso & Hirschmann, 2001), also referred to as island arc ankaramites.

Extent of melt–peridotite interaction

High concentrations of compatible elements (e.g. $Cr_2O_3 > 0{\cdot}4$ wt %) in the Cabo Ortegal pyroxenites suggest that they crystallized from primitive melts, which are known to produce olivine-bearing assemblages at relatively low pressures, if generated in equilibrium with peridotite at higher pressures (Kushiro & Yoder, 1966; O'Hara, 1968; Green & Ringwood, 1969). Apart from very thin layers (<5 cm thick) that may have mechanically included olivine from the host-rock (Fig. 6), olivine-bearing samples are restricted to the most primitive (type 1) pyroxenites (i.e. with the highest Mg# and Cr#). Their high cpx/opx ratios (Fig. 6) point to crystallization along a cpx–olivine cotectic line, as expected from ultra-calcic picrites, but contrasting with the early appearance of opx during the crystallization of boninites (Barsdell & Berry, 1990; Eggins, 1993). These pyroxenites also have lower SiO_2 at similar MgO compared with other types (Fig. 8a) and preserve dunite lenses within them, suggesting that they were produced by incomplete reaction with their dunitic host.

Whereas olivine-free massive (type 2) pyroxenites plot close to the experimental cumulates along the En–CaTs join on the Fo–CaTs–Qz diagram (Fig. 16), type 1 pyroxenites plot close to the Fo–An join within the nepheline-normative, silica-deficient field. This trend between pyroxenites and the peridotite array (i.e. close the Fo apex) is seen in other orogenic pyroxenites, and has been ascribed to melt–peridotite interaction (Bodinier *et al.*, 2008; Lambart *et al.*, 2012). Provided that it occurred at relatively low pressure (<1·2 GPa), such reaction could account for the disappearance of olivine and increasing modal proportion of opx between Cabo Ortegal type 1 and 2 pyroxenites via a peritectic reaction of the type Fo + Si-rich melt = En (e.g. Kelemen, 1986; Kelemen & Ghiorso, 1986; Burg *et al.*, 1998). This is consistent with the preservation of dunite lenses within type 1 pyroxenites, which was interpreted in the Jijal Complex as the result of partial replacement of peridotites (Burg *et al.*, 1998; Garrido *et al.*, 2007; Bodinier *et al.*, 2008). It is also consistent with the small time-integrated amount of metasomatic agent (and/or low melt–rock ratio) inferred from the limited enrichment of LREE and LILE of type 1 pyroxenites. It follows that percolation of residual melts [percolative fractional crystallization of Harte *et al.* (1993)] is likely to be responsible for the metasomatic reactions described above and produced the positive correlation between the amount of layered pyroxenites and the extent of LREE enrichment observed in the host harzburgites (Gravestock, 1992). Such melt-induced metasomatism would also account for the whole-rock budget of Li being dominated by cpx, because Li preferentially partitions into cpx instead of olivine when in the presence of melt (Scambelluri *et al.*, 2006).

Therefore, we conclude that the Cabo Ortegal pyroxenites resulted from melt–peridotite interaction initially involving picritic Ca-rich melts, which produced type 1 pyroxenites (and the protoliths of type 3 pyroxenites), then evolved towards Ca-rich boninitic compositions, crystallizing type 2 pyroxenites. This scenario is supported by the normative compositions of type 1 and 3 pyroxenites trending towards ultra-calcic picrites, and those of type 2 pyroxenites trending towards high-Ca boninites (Fig. 16). In addition, we suggest that the formation of opx-rich (type 4) websterites occurred where boninitic melts interacted with dunites. This is in good agreement with the relatively high NiO in olivine from type 4 pyroxenites reflecting the concentration of Ni in smaller amounts of olivine (Fig. 11d) and the extreme REE patterns measured in their low-modal cpx, as predicted by the chromatographic model.

Source region and melting processes

The high CaO of Cabo Ortegal pyroxenites certainly relates partly to the segregation and accumulation of abundant cpx. However, their high CaO/Al_2O_3 (2·2–11·3) is indicative of a similarly high CaO/Al_2O_3 in their parental melts, which carries important implications with regard to the melting conditions and nature of the source region. In addition, the mineralogical residence sites of the HFSE are one of the major issues in mantle geochemistry (e.g. Haggerty, 1991; Grégoire *et al.*, 2000*a*; Kalfoun *et al.*, 2002), the consistency between the depletion of HFSE in cpx and amphibole and their depletion in whole-rock samples indicates that it must be a primary feature inherited from the composition of the parental melts of the Cabo Ortegal pyroxenites. This a characteristic fingerprint of subduction-related volcanic rocks (e.g. Kelemen *et al.*, 1993; Foley *et al.*, 2000; Münker *et al.*, 2004), although it also has been reported in various other contexts (Salters & Shimizu, 1988; Benoit *et al.*, 1999; Nonnotte *et al.*, 2005). The genesis of parental melts with high CaO/Al_2O_3 for the Cabo Ortegal pyroxenites, of either boninitic or picritic affinities, is therefore considered in this context.

Experimental studies have suggested that boninitic and picritic magmas are generated in equilibrium with refractory sources, representing the residue of earlier melt extraction (Crawford *et al.*, 1989, and references therein). Maximum CaO/Al_2O_3 ratios in such second-stage melts are reached when cpx is consumed from the residue (e.g. Sorbadere *et al.*, 2013), and ideally when earlier (first-stage) melts were derived from a plagioclase lherzolite (i.e. at relatively low pressure; Schmidt *et al.*, 2004). Partial melting of a refractory lherzolite source is consistent with the low cpx/opx source inferred by Gravestock (1992) from the modelling of Ti and Zr depletion during batch melting.

Like present-day boninitic and picritic lavas, the parental melts of the Cabo Ortegal pyroxenites probably acquired their relatively high concentrations of compatible elements such as Ni, Mg and Cr from a depleted mantle source. However, the high LILE/HFSE of both whole-rocks and the main host-rock minerals suggests a metasomatic contribution, in good agreement with the metasomatized harzburgitic source suggested by Gravestock (1992), but contrasting with the primitive-mantle source advocated by Laribi-Halimi (1992). Assuming a mantle-wedge source, this metasomatic contribution may be related to percolation of slab- or mantle-wedge-derived fluids or melts (Meijer, 1980; Hickey & Frey, 1982; Hawkesworth *et al.*, 1993; Tatsumi & Eggins, 1995; Grégoire *et al.*, 2001, 2008). Preferential partitioning of LILE into hydrous fluids or melts extracted from a subducting slab is indeed one of the mechanisms invoked to account for relative depletion of HFSE (e.g. Tatsumi *et al.*, 1986; Saunders *et al.*, 1991; Tatsumi & Eggins, 1995; Kelemen *et al.*, 2003). In addition, the presence of Ti-rich minerals (Foley *et al.*, 2000; Barth *et al.*, 2002; Klemme *et al.*, 2002) or the existence of an amphibole-rich zone (Ionov & Hofmann, 1995) in the mantle wedge, and the stability of chlorite-harzburgite with clinohumite during deserpentinization of the subducting slab (Garrido *et al.*, 2005) have been suggested as ways to retain HFSE where arc magmas are generated and/or transported (Saunders *et al.*, 1980; Green & Pearson, 1987; Ryerson & Watson, 1987).

Alternatively, carbonatitic melts could represent one of the enriched components in the source region. This has been suggested for the genesis of boninitic melts (Falloon & Crawford, 1991) and envisaged by Gravestock (1992) to account for the paradox of high Ca content vs low cpx/opx of the Cabo Ortegal pyroxenites. It is indeed expected from experimental studies that high CO_2/H_2O, as induced by carbonatite metasomatism during partial melting, may shift the melt compositions towards higher normative diopside and therefore higher CaO/Al_2O_3 (Green *et al.*, 2004). Because the partition coefficient for Nb is higher than that for Ta (Green *et al.*, 1992), percolation of carbonatites in the source region of their parental melts could also account for the high Nb/Ta of Cabo Ortegal pyroxenites [$(Nb/Ta)_{PM} = 0{\cdot}9$–$3{\cdot}1$]. However, the presence of rutile in the source region could also have controlled the amount of Ta entering the melt fraction (Foley *et al.*, 2000; Rapp *et al.*, 2003; Klemme *et al.*, 2005).

Tectonic and geodynamic significance

There are relatively robust constraints on the geodynamic setting associated with the metamorphic complexes of northwestern Iberia (e.g. Martínez Catalán *et al.*, 2009; Arenas *et al.*, 2014*a*). However, the origin of some of the high-pressure allochthonous units is uncertain. In the Cabo Ortegal Complex, the ultramafic massifs are structurally associated with granulites, and the nature of their protoliths is debated. They have been variously interpreted as deep residues of anatexis (Drury, 1980), a stratiform gabbroic complex at the base of the continental crust (Galán & Marcos, 1997) or as mafic rocks related to an island arc formed near a continental margin (Peucat *et al.*, 1990). These granulites have also been interpreted as the products of early retrogression during exhumation of the eclogites (Galán & Marcos, 2000). Considering their lithological diversity, ranging from mainly mafic and ultramafic, to intermediate and gneissic (Ábalos *et al.*, 2003), these interpretations are not mutually exclusive. Nonetheless, for the purpose of the following discussion, we assume that the Cabo Ortegal granulites represent remnants of a volcanic-arc crust, in good agreement with the volcano-sedimentary protolith inferred for the associated HP gneisses (Gil Ibarguchi *et al.*, 1990).

Formation of the Cabo Ortegal pyroxenites

Assuming a sub-arc environment, the Cabo Ortegal pyroxenites may represent layered cumulates, along with the associated dunites ± chromitites (Moreno *et al.*, 2001). This has been documented in ophiolites, where ultramafic cumulates may reach several hundred meters in thickness (e.g. Bédard & Hébert, 1996; Ceuleneer & Le Sueur, 2008; Clénet *et al.*, 2010). However, we are aware of only very limited evidence of magmatic stratigraphy [e.g. rare variations of modal proportions across layers, as reported in this study (Fig. 14a) and by Girardeau & Gil Ibarguchi (1991)]. The increasing content of platinum-group elements (PGE) and Cr# in chromitites exposed throughout the western cliffs (Moreno, 1999; Moreno *et al.*, 2001) has also been suggested as indicative of magmatic stratigraphy, but it is constrained by the arguable existence of the Trans-Herbeira Fault (THF), as discussed above. In addition, although the lithological stratigraphy of the Herbeira massif could be compared with a cumulate series, such as that of the suprasubduction-zone Trinity ophiolite (Ceuleneer & Le Sueur, 2008), it is also comparable with the transition from the peridotite to the pyroxenite zones of the Jijal ultramafic section in the Kohistan complex, where the previously proposed cumulative origin (Khan *et al.*, 1993) has been strongly questioned. The preservation of dunite 'flames, streaks and wisps' in websterites of the peridotite zone is now indeed regarded as resulting from partial replacement of peridotite following melt–rock interaction at decreasing melt–rock ratio close to the peridotite solidus (Burg *et al.*, 1998; Garrido *et al.*, 2007; Bodinier *et al.*, 2008). These are comparable with the Cabo Ortegal type 1 pyroxenites whereas the final products of reaction at higher melt–rock ratio, represented by olivine-free websterites in the pyroxenite zone, are similar to type 2 pyroxenites. They may be related to the transition from channelled porous flows to dykes and veins (e.g. Bodinier *et al.*, 2008), suggesting the existence of a thermally controlled permeability barrier, which in turn may be indicative of regional tectonic processes (e.g. Van der Wal & Bodinier, 1996; Burg *et al.*, 1998; Garrido & Bodinier, 1999; Python & Ceuleneer, 2003; Rabinowicz & Ceuleneer, 2005; Bouilhol *et al.*, 2011). Significant modification of melt distribution may also be envisaged if a carbonatite component was involved in the source region, as experiments show that degassing of C–H–O fluids at $<$ 1 GPa should increase melt viscosity (Green *et al.*, 2004). However, the strong deformational overprint resulting in the transposition of lithological contacts hinders further interpretation.

Generation of the parental melts

Few constraints are available regarding the tectonic setting associated with the generation of the parental melts of Cabo Ortegal pyroxenites. The requirement for hydrous $\pm$ carbonatitic components contributing to the flux melting of a refractory mantle source seems reasonable in a mantle-wedge environment (Green *et al.*, 2004). Generation of boninitic or picritic melts also implies that the melting region was relatively shallow ($<$2 GPa), or that the melts re-equilibrated at shallow depth, although this might not be correct if a significant volume of cpx-rich lithologies was involved in the melting (Schmidt *et al.*, 2004, and references therein; Sorbadere *et al.*, 2011, 2013). In addition, the requirement for high temperatures (1150–1350°C) at shallow depth may have tectonic implications, which are beyond the scope of this study.

Prograde and retrograde metamorphism

The occurrence of garnet as relatively undeformed coronas around spinel (Figs 4c and 5a) is indicative of syn- to post-kinematic prograde metamorphism. Previous thermobarometric calculations put the re-equilibration of the mineral assemblage at 1·6–1·8 GPa and 800°C (Girardeau & Gil Ibarguchi, 1991), in good agreement with calculations made from the data presented here (1·7 GPa and 780°C; Supplementary Data Electronic Appendix 3). We suggest that this episode corresponds to the delamination of the arc root as a consequence of its negative buoyancy (Müntener *et al.*, 2001), followed by its incorporation into a subduction zone. The Cabo Ortegal Complex may thus represent a unique example of a delaminated arc root, a 'Herbeira delaminated arc root', following Moreno *et al.* (2001).

Previous structural studies and kinematic reconstructions have revealed that the ultramafic massifs of the Cabo Ortegal Complex experienced an HP–HT metamorphic event and were reorganized and exhumed as a single assemblage with eclogites, granulites and HP gneisses (Ábalos *et al.*, 2003). The chronology of these tectonothermal events remains controversial and potentially implies exhumation over more than one orogenic cycle (Marcos *et al.*, 2002). However, the combination of subduction-channel processes followed by orogenic deformation remains a reasonable explanation for the retrograde metamorphism (and hydration) of the pyroxenites and of their host peridotites, resulting in amphibolitization and serpentinization. Within this scenario, the dominantly interstitial position of base-metal sulfides and their common association with amphibole (Fig. 3e) suggest that addition of S probably accompanied this subduction-related episode and is thus unrelated to the main magmatic or metasomatic episode. Indeed, the abundance of sulfides (up 1 wt %) cannot be accounted for by the low S content of boninitic or picritic melts (e.g. Cooper *et al.*, 2010; Ripley & Li, 2013).

CONCLUSIONS

Unusually abundant pyroxenites occur interlayered with dunites within the harzburgites of the Herbeira massif in the Cabo Ortegal Complex, Spain, which is part of the Variscan suture in northwestern Iberia. We report whole-rock and *in situ* major and trace element compositions of 30 samples, including pyroxenites with dunite lenses (type 1), massive (type 2) and foliated (type 3) pyroxenites, and subordinate opx-rich pyroxenites (type 4). The field- and petrography-based classification is consistent with the composition of minerals and whole-rocks, which exhibit a range of REE patterns from spoon-shaped to LREE-enriched and LREE-to-MREE fractionated. The Cabo Ortegal pyroxenites have very high CaO/Al_2O_3 ratios, high SiO_2 and Cr_2O_3, low TiO_2 and alkalis, and ubiquitous LILE/HFSE enrichment. These characteristics reflect a combination of magmatic differentiation and melt–peridotite interaction, which may have occurred in a sub-arc environment prior to the intrusion of the massif into a subduction zone according to a six-step scenario.

1. Generation of primitive hydrous melts similar to ultra-calcic picrites (i.e. with notably low Al/Si and high Ca/Al), through relatively shallow ($<$2 GPa), low-degree flux melting of a refractory lherzolitic source that previously had experienced melt extraction at shallower depth and/or refertilization via carbonatite metasomatism.
2. Intrusion of the melts at higher levels inducing melt–peridotite interaction and crystal segregation at decreasing melt–rock ratio, thus partially replacing host-rocks still preserved in type 1 (and in the protoliths of type 3) pyroxenites. Dominantly harzburgitic lithologies and the occurrence of strongly deformed thinly layered pyroxenites suggest that this region also previously experienced melt extraction.
3. Formation of massive (type 2) pyroxenites from boninitic melts at higher melt–rock ratios, potentially as veins or dykes and/or following the completion of a peritectic reaction (Fo + Si-rich melt = En). We suggest that these Si-rich melts may have been produced by differentiation of the initially picritic melts, which notably produced opx-rich (type 4) pyroxenites upon interaction with dunites.

4. Chromatographic re-equilibration between pyroxenites (and their host peridotites) and percolating or trapped residual melts. This episode may have been accompanied by late-magmatic or metasomatic precipitation of amphibole (percolative fractional crystallization).
5. Development of isoclinal folds and of a high-temperature tectonic foliation followed by the development of sheath folds and locally mylonite during high-shear strain deformation. This transition accompanied by prograde metamorphism under eclogite-facies conditions may represent the delamination of an arc root and its incorporation into a subduction zone. Foliated (type 3) pyroxenites were produced during this episode, preferentially at the expense of dunite lenses from the protoliths of type 1 pyroxenites.
6. Amphibolitization of the pyroxenites and serpentinization of their host peridotites during retrograde metamorphism and hydration. This episode is particularly recorded in type 3 pyroxenites and may relate to syn-subduction exhumation.

On the basis of this scenario, we suggest that the Cabo Ortegal Complex preserves a 'Herbeira delaminated arc root' bearing evidence for the significant role of melt–peridotite interaction during the differentiation of primitive arc magmas at depth.

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SUPPLEMENTARY DATA

Supplementary data for this paper are available at *Journal of Petrology* online.

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